

NexStar 2026 Melbourne - Chengdu Sprint

CDEG - Group A

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Executive Summary

Chengdu Environment Group (CDEG) asked how it can enter Australia's waste-to-energy (WtE) and water markets, starting with Victoria. This report answers from CDEG's own materials, public sources and our analysis. We did no interviews at this stage.

Key findings

- **Victoria needs operators, not new licences.** The 2.5 Mt/yr thermal WtE cap is 94% allocated: 2.35 Mt to seven licences in August 2025 [1]. None of the licensed plants has reached construction, and they will need builders and operators around 2029–31.
- **CDEG exports operating skill, not its business model.** In China, WtE earns from government concessions and subsidised power. In Victoria, private owners earn gate fees that compete with a $177.19/t landfill levy [2]. CDEG runs 12,000 t/d of grate capacity now that Wanxing III is on the grid [3, 4], but it buys its equipment.
- **Four licence holders are big enough for CDEG's 500 t/d minimum line:** Cleanaway (760,000 t/yr), HiQ (750,000 t/yr), Melbourne Water (450,000 t/yr) and REL Laverton (280,000 t/yr) [1, 5].
- **One O&M contract is worth $27–39m a year.** At an assumed $87/t, the licensed plants would spend about $200m a year on operations once built, of which about $65–130m is contestable once political risk and self-operating owners are taken out [1] [Assumption]. One contract would earn CDEG $2.7–5.9m a year in margin from about 2030–31. Entry costs about $9–10m over three years, which one contract repays in two to four years of operation.
- **The numbers work only at high utilisation.** A plant with two 500 t/d lines, built at the Australian benchmark of about $1,600 per tonne of yearly capacity, needs a gate fee of about $205/t at 85% utilisation, below the metro landfill price of about $247/t. At 70% utilisation it needs $253/t and does not compete. A 25% lower build cost would cut this to about $149/t, but an overrun like Kwinana's would make the plant uncompetitive even with that saving. Victoria's grant programs exclude thermal WtE, and a possible WtE levy and the 28 November 2026 state election add policy risk [6, 7].
- **Social licence is the main risk.** Of 537 community submissions to the parliamentary inquiry, 86% raise health and air pollution and 56% name Sunbury or Bulla, HiQ's site [8]. Only 4 express concern about foreign ownership.
- **Water and organics are the fastest start.** Biosolids and food organics need no cap licence: 167,000 t and 439,000 t went to disposal in 2024–25 [9, 10]. In meters the field is crowded: Melbourne's two largest digital-meter programs are contracted, and three Chinese makers already hold Australian approvals [11, 12]. Regional water corporations and the 2028 Water Price Review are the opening [13].
- **Chinese entry works best as a service operator.** BEWG's TRILITY runs Australian water plants under local management and, with CCCC-owned John Holland, won Sydney Water's Upper South Creek plant. Control of critical networks has been refused, and Cleanaway, the largest licence holder, is being bought by EQT.

Recommendations

1. **Lead with O&M and technical services** for the four large licence holders, entered through a local consortium, for example with TRILITY or John Holland. Expect the first O&M revenue around 2030–31, later if plants slip.
2. **Start now with low-capital moves:** a digital-meter pilot with a regional water corporation, a biosolids co-processing proposal to Melbourne Water, and joint R&D (CCUS, sludge) with a Victorian university.
3. **Keep technology introduction, co-investment and coal-site conversion as a later track,** opened only once policy, regulatory and approval risks are clear. Avoid firm commitments until after the 28 November 2026 state election and the government's response to the inquiry.

First-year actions: verify Wanxing and Longfeng emissions in EU format; prepare an English technical dossier; start NMI pattern approval for one meter; apply to FIRB and complete the China-side outbound filing to set up a Melbourne entity, which an SOE needs for any new Australian business [14]; sign two MoUs (a consortium partner and a university).

1.0 Introduction

1.1 About CDEG

Chengdu Environment Investment Group is a state-owned enterprise (SOE) owned by the City of Chengdu. It has 135 subsidiaries and about 8,200 staff, and holds a domestic AAA credit rating [15, 16]. It controls the listed Xingrong Environment (000598.SZ), whose 2025 revenue was RMB 9.07bn.

Xingrong runs about 4.4 Mt/d of water supply, 5.0 Mt/d of wastewater treatment and 1.3 Mt/d of reclaimed water, plus 3,116 t/d of sludge and 1,950 t/d of food-waste treatment [3]. Its four WtE plants, Wanxing I, II and III and Longfeng, total 12,000 t/d [3]. The wider group also transfers 7,230 t/d of waste and treats about 150,000 t/yr of hazardous waste [15].

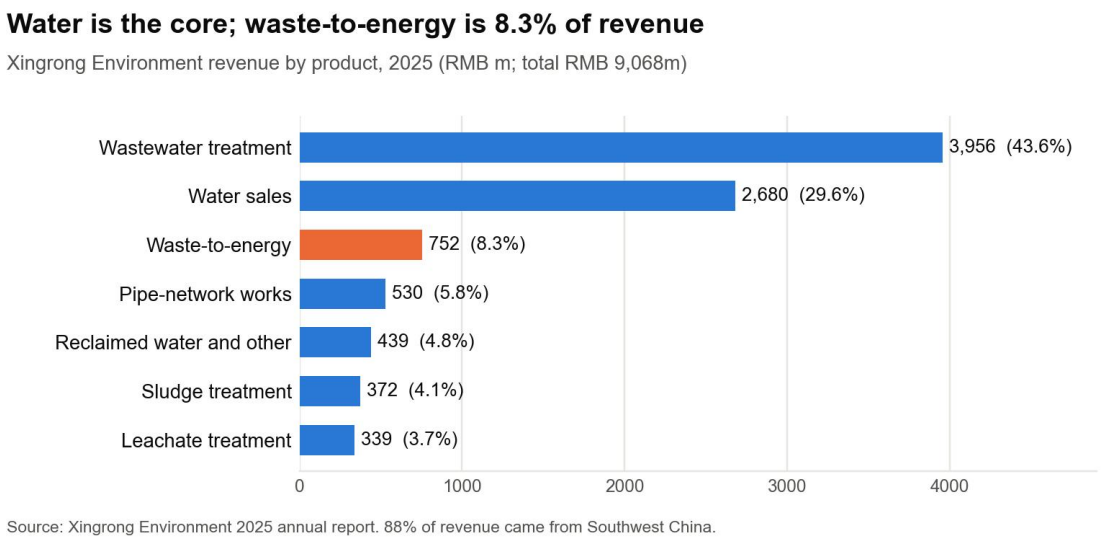


Figure 1. Xingrong Environment revenue by product, 2025

Water is the core business. WtE earned RMB 752m in 2025, 8.3% of revenue, at a 45.6% gross margin, and sold 1,020 GWh of power [3]. 88% of Xingrong's revenue came from Southwest China, and its only overseas holding is an associate in Lahore [3]. Therefore, CDEG has deep operating experience, but almost all of it is in China.

1.2 What CDEG can and cannot export

CDEG's exportable strength is operating skill. It ran about 6,900 t/d of grate WtE before Wanxing III [3]. Xingrong announced on 7 May 2026 that Wanxing III had completed its 72+24-hour full-load trial [4, 17]. By August 2026 the plant was feeding the grid stably, and its tipping fee was still being set [4]. Its furnaces co-fire 10% sludge, and Wanxing III is designed for 800 t/d of sludge and 800 t/d of food waste [5, 15]. Its flue-gas train (SNCR, semi-dry and dry scrubbing, activated carbon, baghouse and SCR) is built to EU-style limits, and its daily emissions are published [18, 19].

Three other assets travel with conditions. The food-waste equipment was built for wet, high-impurity Chinese food waste, so it suits commercial food waste better than mixed FOGO [15] **[Inferred]**. The NB-IoT ultrasonic meters (Class 2, R160–R400) and drainage sensors are commercial in Chengdu and four other cities, but need NMI approval before trade use in Australia [16]. The CCUS line is still at research stage, which fits joint R&D.

Two things do not travel. China's concession-plus-subsidised-power model has no equivalent in Victoria. And CDEG does not own furnace technology: it buys its equipment.

1.3 CDEG's questions

CDEG has some concerns about entering the market. The key questions they care about are

as follows.

1. Which rules govern market access, emissions and grid connection? (2.3; Appendix C)

2. What waste does Australia have, and does it suit CDEG's grates and co-processing? (2.2; 3.3)
3. Where are the WtE projects to operate, upgrade or invest in? (3.1; 4.1; 6.1)
4. Who buys smart water meters, and through which channels? (3.1; 4.3; 6.4)
5. Which research partners suit joint R&D on sludge, CCUS and AI for water? (3.1; 6.4)
6. Who pays, and how big is the market? (3.2; 3.4)
7. Who are the competitors, and how have Chinese firms fared in Australia? (5.0)

1.4 Method

We cover Victoria first and compare NSW, WA, QLD and SA (Appendix B). We use three kinds of evidence: CDEG's own materials, public sources (government, regulator and parliament records, company filings and industry reports) and our own analysis. Due to the time limitation, we did no interviews at this stage. Key figures carry a reference number and, where useful, a label: [Official], [Industry], [Company], [Inferred], [Assumption] or [To verify] (Appendix G).

2.0 Market Context

2.1 Victoria's problem and opportunity

Victoria makes more residual waste than its WtE cap can take, landfill is getting dearer every year, and the licensed plants will need operators from about 2029.

Residual waste outgrows the cap. Victoria generated 14.6 Mt of waste in 2023 and recovered 69%. That rate has been flat since 2016–17 [20] [Official]. Residual waste was 5.7 Mt in 2024 and is projected to reach 8.9 Mt by 2053. The state is not on track for its 80% diversion target by 2030, and Melbourne expects to run out of landfill space in the mid-2030s [20, 6] [Official]. Thermal WtE is capped at 2.5 Mt/yr, and 2.35 Mt (94%) was allocated on 17 August 2025 [1, 6]. The Head of Recycling Victoria told the parliamentary inquiry that no new capacity can be allocated above the cap, and existing operators hold separate licences, which takes total licensed capacity to just over 3.5 Mt/yr [6]. So even if every licensed plant is built, more than 3 Mt/yr of residual waste still needs another route. That holds even if FOGO cuts residual tonnes by a third (3.3) [Inferred].

Landfill is priced to lose. The metro municipal and industrial levy rose from $129.27/t in 2023–24 to $132.76/t in 2024–25, $169.79/t in 2025–26 and $177.19/t in 2026–27, a 33% rise in two years, when it moved to fee units (10.26 units for metro waste) [2]. NSW's metro levy was $174.20/t in 2025–26 [21]. Landfill operators charge their own gate fee on top. Gate fees are not published consistently. We assume about $70/t, which puts the metro landfill price that WtE must beat at about $247/t [Assumption]. Cleanaway's posted putrescible rate at its Melbourne Regional Landfill, about $448/t including GST, is an upper bound [22]. Contracted council rates are lower and unpublished. A WtE plant pays no levy on the waste it burns, although levy applies to any ash it sends to landfill. It can set its gate fee just under the landfill price and still win the waste.

The plants are licensed but not built. None of the seven cap-licence projects has reached construction. The four big enough for a CDEG line plan to start between about 2029 and 2031 (Appendix A). Cleanaway withdrew its EPA application in September 2025 to re-file at 760,000 t/yr [23], and its takeover by EQT is due to complete by March 2027 [24]. Each of these owners will need a builder, a technology supplier and an operator. Operating is the role CDEG wants (1.2; 5.1).

Three streams sit outside the cap. About 1.6 Mt of organics a year still goes to landfill. Every household must have food and garden organics (FOGO) collection by 2030, and the standard four-bin system is due by 1 July 2027, although that deadline was not yet legislated in March 2026 [20, 25, 26]. In 2024–25, 439,000 t of food organics and 167,000 t of biosolids went to disposal [9] [Official]. Biosolids are "exempt waste": burning them alone needs no cap licence [10]. CDEG already treats all three streams in Chengdu (1.2).

Water has its own timetable. The large metropolitan meter programs are already contracted: South East Water had installed 200,000 digital meters by January 2026 [11]. For regional water corporations, the next budget window is the 2028 Water Price Review, with submissions due 13 September 2027 [13, 27] (4.3).

Table 1. Victoria's residual-waste gap

Measure	Mt/yr	Source
Residual waste, 2024	5.7	[20] [Official]
Residual waste, 2053 projection	8.9	[20] [Official]
Thermal WtE cap	2.5	[28, 6]
Cap allocated (seven licences, Aug 2025)	2.35	[1]
Allocated to the four licences large enough for a 500 t/d line	2.24	[1] (Appendix A)
Residual not covered by allocated cap, 2024	3.35	[Inferred]

Sources: [1, 28, 6, 20]. The last row is 5.7 Mt minus the 2.35 Mt allocated. The 14.6 Mt and 69% figures are for 2023 and use a different definition, so they do not reconcile exactly with 5.7 Mt.

So what for CDEG. The shortage is of operators, not of waste or licences. CDEG cannot win a cap licence, but it can sell the operating skill the licence holders will need around 2029–31. Meanwhile it can start in organics and biosolids, which need no cap.

2.2 China’s WtE model vs Victoria

In China, local government makes the decisions.

- **Waste supply**: It grants a 25–30-year BOT or BOO concession, now generally up to 40 years [3, 29], and pays a tipping fee per tonne [3, 30].
- **Main revenue**: Electricity earns about two-thirds of revenue: the first 280 kWh per tonne of waste earns a benchmark RMB 0.65/kWh, about RMB 182/t, and the rest earns the local coal-power price [30, 31]. The tipping fee, about RMB 100/t, provides the other third [30]. Plants connected since September 2020 share this subsidy with local government, and the central share will be phased out [32].
- **Emission**: Since 2020, automatic monitoring data have been usable as evidence of violations [33], and plants publish their emissions data on a public MEE platform [19, 34].
- **Feedstock and siting**: CDEG’s plants do no front-end sorting, and its siting rule of thumb, within 50 km of the city, comes from this setting [5].

In Victoria, a private developer must win a cap licence, EPA development and operating licences, and planning approval [28].

- **Waste supply**: Council consortia sign long supply contracts: nine south-east metropolitan councils agreed a 25-year waste supply agreement in 2024 [35].
- **Main revenue**: There are no large-scale thermal WtE facilities operating in Victoria at present, so the national estimate is used [36]. Australian estimates put gate fees at about 60% of revenue, electricity at about 25% and recovered metals and other recyclables at about 15% [37]. In Harford’s 300,000 t/yr model plant, a gate fee of $150–250/t earns $45–75m a year, 241,920 MWh of electricity at $90/MWh earns about $22m ($73/t) and other products $16m ($53/t) [37]. **Victoria's own modelling assumes a gate fee of $237/t and electricity at $80.11/MWh [36].**
- **Emission**: EPA expects best practice in siting, design and emission control [38]; the Maryvale approval even ties mercury monitoring to European Commission best available techniques [39].
- **Feedstock and siting**: Only waste that cannot be recycled may be burnt [40]. Sites are decided through planning, and they are strongly contested (4.4).

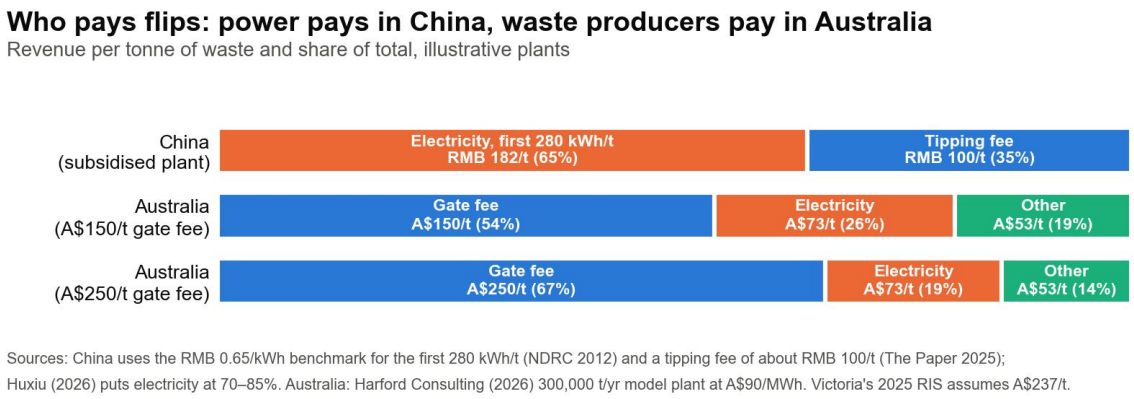


Figure 2. WtE revenue per tonne in China and Australia

Why CDEG needs going abroad. China's incineration capacity reached about 1.18 Mt/d at the end of 2025, against a 14th Five-Year Plan target of about 0.8 Mt/d [41, 42]. Estimates of plant utilisation differ: about 60% on an industry estimate [30], against an average load rate of 88.7% in the official count for 2024 [43]. Power subsidies end after 82,500 subsidised hours or 15 years from grid connection [44], so tipping fees have risen from about RMB 60–70/t to about RMB 100/t, and up to RMB 150/t in some cities [30]. Chinese firms had 79 overseas WtE projects by May 2025 [30] and joined 12 more, with over 16,200 t/d of capacity, in the first half of 2026 [45]. CDEG's own Pakistan project (Lahore, 40 MW, MoU in 2017) and its Uzbekistan project stopped for policy reasons [46, 5].

Why the waste differs. Chinese household waste is wet, with a lower heating value (LHV) of about 3–6.7 MJ/kg; European residual waste is 9–11 MJ/kg, on a 2017 industry estimate [47]. Once FOGO takes food out, Victorian residual waste should be drier and closer to European values [Inferred]. The question for CDEG is therefore high-LHV fit, not dewatering (3.3; Appendix C).

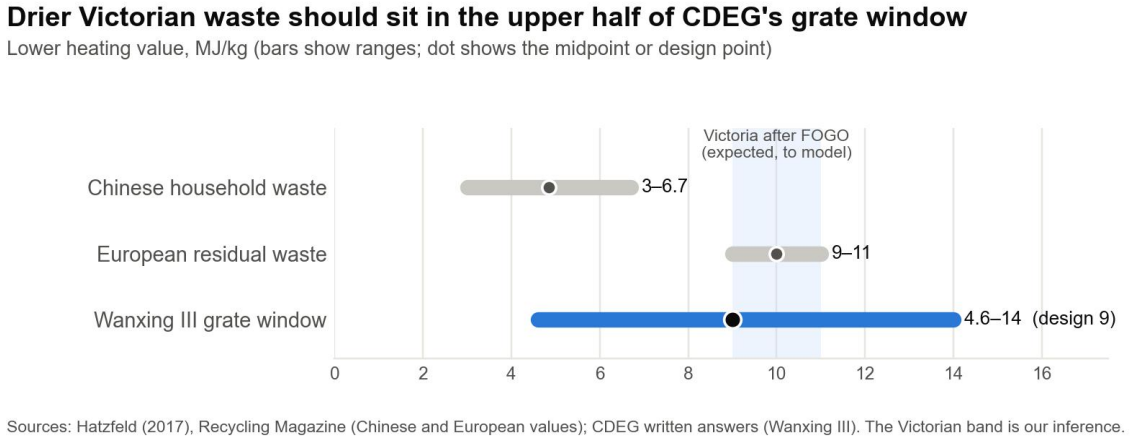


Figure 3. Heating value of Chinese and European waste against CDEG's grate window

2.3 Rules a project must meet

A Victorian WtE plant needs four separate approvals. A Chinese SOE needs approvals in both countries. That is why CDEG should enter as a contractor to a licence holder, not as an owner.

From mid-2026, Recycling Victoria became EPA Victoria's Circular Economy division, so one regulator now handles both the cap licence and the environmental licences [28, 22] [To verify] exact date (team files say 30 June or 1 July 2026). The owner, not CDEG, holds every project approval in Table 2. CDEG's own approvals are the corporate ones at the bottom of the table.

Table 2. Approvals for a thermal WtE plant in Victoria, and who holds them

Approval	Body and law	What it tests	Who holds it	What it means for CDEG

Approval	Body and law	What it tests	Who holds it	What it means for CDEG
Cap licence	EPA Circular Economy division; *Circular Economy (Waste Reduction and Recycling) Act 2021*	Nearness to feedstock and offtake, viability, permits held, jobs, energy output, fit and proper person, climate impact [28, 22]	Owner	94% allocated, so CDEG enters through a holder [1]
EPA development licence, then operating licence	EPA Victoria; *Environment Protection Act 2017*; Energy from waste guideline 1559.1	Best available techniques. In practice EU BAT is the reference (Appendix C) [38]	Owner; operator named in the application	CDEG's operating and emissions record supports the owner's case
Planning permit (possibly an EES)	Council or Minister for Planning	Siting, buffers, traffic, community. There is no minimum buffer distance [6] (Finding 46)	Owner	The main social-licence hurdle (4.4)
Federal environment referral	DCCEEW; *EPBC Act*	Matters of national environmental significance, if triggered	Owner	Site-specific
Grid connection	AEMO and the network owner	Connection studies and queue position	Owner	Easier at former coal sites (3.1e)
Permitted-waste rules	WtE Scheme scope guidance	No recyclables. Deposit drink containers are banned in any amount. Commercial waste must pass the "not practicable to recycle" (TEEP) test [10]	Owner and operator	Needs sorting and tipping-floor checks before the furnace (3.3)
Bottom-ash reuse	EPA Victoria, case by case	Tests before ash is used as aggregate [6] (Findings 5–7)	Owner and operator	Matches CDEG's 100% slag reuse in Chengdu [18]
Foreign investment	FIRB; *Foreign Acquisitions and Takeovers Act 1975*	An SOE is a foreign government investor, so the threshold is $0. A new Australian subsidiary needs approval [14, 48]	CDEG	Service contracts generally do not trigger FIRB; equity does [To verify] with a lawyer
Critical infrastructure	Home Affairs; *SOCI Act 2018*	Risk-management programs that flow down to contractors [49]	Asset owner, passed on to contractors	CDEG must pass the owner's supplier and personnel checks
China-side outbound approvals	NDRC (Order No. 49, effective 1 Dec 2026), MOFCOM, SAFE, Chengdu SASAC [50]	Outbound investment and foreign-exchange filing	CDEG	Needed for a Melbourne entity and for any parent guarantee

Sources: [1, 14, 18, 28, 38, 49, 48, 50, 6, 10, 22]. Emission limits are compared in Appendix C.

The rules are tightening. The parliamentary inquiry (August 2026) made 59 findings and 33 recommendations [6]. Four matter most for an operator:

- continuous emissions monitoring with real-time public reporting (Recommendation 22)
- international best-practice standards written into regulation (Recommendation 19)
- guidance to keep council supply contracts flexible (Recommendation 12)
- a review of whether economic incentives still favour recycling over thermal treatment (Recommendation 13). That could mean a WtE levy.

The government has six months to respond, so a response is due around February 2027 [6]. The state election on 28 November 2026 comes first [51].

So what for CDEG. As a contractor, CDEG needs only the corporate approvals: FIRB for a Melbourne subsidiary (not for a service contract alone, subject to legal advice), the China-side filings and SOCI clearance. It carries none of the project-approval risk. In return, its emissions data must meet the bar the inquiry has set.

3.0 Market Opportunity and Commercial Viability

3.1 Target markets

Four licence holders are large enough for a CDEG line, and up to three of them could use grate technology. Biosolids and organics sit outside the cap. A former coal site in the Latrobe Valley is a later option, not a first step.

We see five segments: (a) WtE O&M for licence holders, (b) organics, sludge and biosolids outside the cap, (c) water digital products, (d) R&D partners and (e) coal-site conversion.

(a) WtE O&M for licence holders

CDEG's minimum economic line is 500 t/d. Below 250 t/d is too small [5]. A 500 t/d line handles about 155,000–182,500 t/yr at 85–100% availability. CDEG's written answer adds that a project should have at least two lines (1,000–1,500 t/d) so the plant keeps running while one line is down [18]. Four licence holders pass the line test, but Recovered Energy Laverton (about 770 t/d) falls below the two-line project minimum and uses gasification (Table 3). The other three cap licences (Knox, Zerogen, Liquid Power) are each smaller than one 250 t/d line [1, 5].

Table 3. Licence holders large enough for a CDEG line

Holder (site)	Cap t/yr	500 t/d lines	Technology	Status and operator	Fit for CDEG
Cleanaway (Wollert)	760,000	4–5	Not announced; open technology EOI [52]	EPA application to be re-filed; EQT takeover to complete by Mar 2027 [23, 24]	High, if the new owner does not self-operate
HiQ (Bulla, near Sunbury)	750,000	4–5	Grate, Kanadevia Inova (App. A)	No planning approval; strongest local opposition (4.4); Opposition pledged to scrap it [53]	High technically; highest political risk
Melbourne Water (ETP or WTP)	450,000	2–3	Open; waste-heat sludge drying studied [70, App. B]	Feasibility 2026–31; private participation possible [6]; biosolids EOI and tender, Mar–Jun 2025, minimum 70,000 t reported [54, 22]. Purpose of the cap licence not fully disclosed	Highest: CDEG co-fires sludge at Wanxing [18]
Recovered Energy Laverton	280,000	1–2	Gasification (App. A)	Holds both licences; no start date or operator [55]	Low–medium: CDEG has no confirmed gasification experience [5]

Sources: [1, 5, 18, 23, 24, 55, 6, 54, 52, 53]; line counts are our calculation from the 500 t/d minimum at nameplate. At Victorian heating values a line takes fewer tonnes (3.3), which adds about one line to the Cleanaway and HiQ counts [Inferred]. Maryvale is not a target: its operator (Veolia) is inside the consortium (Appendix A).

(b) Organics, sludge and biosolids outside the cap

- **Biosolids.** Victoria produced 366,000 t of biosolids in 2024–25, and 167,000 t went to disposal [9]. Biosolids burnt on their own need no cap licence [10]. Water corporations buy sludge services through long contracts. For example, Barwon Water's biosolids PPP runs until about 2032 and is operated by BEWG-owned TRILITY [22] [To verify]. Melbourne Water ran a "biosolids processing solution" tender from March to June 2025 [54]. CDEG's Wanxing III is designed to co-fire 800 t/d of sludge (Appendix D).

- **Food organics.** 439,000 t went to disposal in 2024–25 [9]. FOGO for all households by 2030 will raise collection, but contamination is a known gap: the state names decontamination and pre-processing as investment needs [20]. CDEG runs 1,950 t/d of food-waste digestion in China, using equipment built for wet, high-impurity waste (1.2).

Table 4. Outside-the-cap streams

Stream	To disposal, 2024–25	Buyers	Incumbents	Timing	What CDEG sells
Biosolids	167,000 t	Melbourne Water; regional water corporations	TRILITY (Barwon); RMIT PYROCO pyrolysis [22]	Melbourne Water feasibility 2026–31; Barwon re-tender ~2032	Sludge drying and incineration O&M or technical services
Food organics	439,000 t	Councils; organics processors (e.g. Repurpose It)	80 organics facilities, 8 digesters [20]	FOGO for all households by 2030	Decontamination and digestion equipment, plus a pilot

Sources: [20, 9, 54, 22]. Prices per tonne are not published; we size these streams in tonnes only.

Both streams need no cap licence and no equity. A pilot or service contract may still need the Melbourne entity, and therefore FIRB (2.3). They are the near-term half of the two-speed entry (6.1, 6.5).

(c) Water digital products

The ESC regulates 16 water businesses, whose 2028–33 spending is set by June 2028 [13]. CDEG's NB-IoT meters and drainage sensors target this cycle.

(d) R&D partners

RMIT's PYROCO biosolids pyrolysis already runs an A$11m demonstration plant with water corporations [56]. CDEG's CCUS, sludge and AI-for-water work suits joint projects.

(e) Coal-site conversion and siting on Australian conditions

CDEG's rule of thumb in China is that a plant is generally no more than about 50 km from the city, because haul cost decides viability [5]. It is not a hard rule here. Victoria's levy changes that rule. A plant here competes with a ~$247/t landfill price, not a government-set tipping fee. In our base case the plant breaks even at $205/t, which leaves about $42/t of headroom for longer haul, before the items the model leaves out (3.4) [Inferred].

- **Latrobe Valley sites.** These are Hazelwood (closed 2017; its rehabilitation EES was on exhibition to 2 November 2026, with a hearing in February 2027), Yallourn (closing mid-2028) and Loy Yang A (closing 2035) [57, 58, 59]. They offer industrial land, terminal stations and power-station skills. Maryvale EfW shows the valley can win approval [60].
- **Haul cost.** Latrobe is about 150 km from Melbourne. At 2016 BITRE freight rates plus our handling assumptions, that adds about $6–19/t compared with a metro site, or $3–10m a year on 500,000 t [61, 22] [Assumption]. Waste is bulky, so real quotes could be higher. Rail is not clearly cheaper than road over 150 km. Its case is fewer truck trips and spare paths, as on Opal's planned waste train to Maryvale [62].
- **Verdict.** In the base case the extra haul fits inside the headroom, but at 70% utilisation there is no headroom at all. A Latrobe hub therefore works only for a licence holder whose metro site is blocked, and only at high utilisation. We score it in 6.1 as a later-stage option. In the siting screen (Model 3, Appendix E), Melbourne Water's two treatment plants and Laverton North tie for first. Transfer stations, water and buffers are scored there. Because an asset-light CDEG does not choose the site, the screen is a tool for advising a licence holder.

3.2 Market size: TAM, SAM and SOM

Once built, Victoria's licensed WtE plants would spend about $200m a year on operations. After removing the most contested plant and owners likely to self-operate, about $65–130m is contestable. One contract would bring CDEG $27–39m a year in revenue from about 2030–31.

Formula. Market value = tonnes processed × O&M contract value per tonne. The contract value is what an owner pays an operator: labour, maintenance, consumables, management and the operator's margin. Reagents and ash disposal may be passed through, which is one reason for the wide range. Gate fees, power sales and capital belong to the owner. The values are annual and start only once plants operate.

O&M value per tonne. We take about $87/t as the base case, from three cross-checks:

- our cost model: fixed O&M at 2.5% of capex plus $40/t of variable costs, at 85% utilisation (Appendix E) [Assumption]
- $70/t opex default in the team website model (wte-payback) [22]
- Metro Vancouver's contractor-operated plant: C$113–138/t in 2023–24 (about A$124–152). That figure includes ash disposal and debt service. The operating-only cost is about C$93–112/t, or roughly A$100–125 [63] [Industry]

Our range is $60–110/t. Two of these checks are our own models, so the figure is an assumption until CDEG gives its operating cost per tonne and an Australian benchmark is found [To verify].

Table 5. WtE O&M market in Victoria

Layer	Definition	Tonnes/yr	Value at $87/t	Range ($60–110/t)
TAM	All cap-licensed thermal tonnes	2.35 Mt	$204m	$141–258m
SAM	The four licences with ≥ one 500 t/d line	2.24 Mt	$195m	$134–246m
SAM (excluding gasification)	Cleanaway, HiQ, Melbourne Water	1.96 Mt	$171m	$118–216m
Risk-adjusted SAM	Above, without HiQ (political risk); half again if owners self-operate or take vendor-bundled O&M	0.75–1.49 Mt	$65–130m	$45–164m
SOM (scenario)	One O&M contract by about 2031: a two-line stage (310,000 t) or Melbourne Water (450,000 t)	0.31–0.45 Mt	$27–39m	$19–50m

Sources: tonnes [1]; $/t triangulated from [63, 64, 22]. Values are annual and in 2026 dollars, earned only once plants operate (about 2029–31) [Inferred].

Reading the numbers. TAM and SAM are close because almost all cap tonnes sit with the four large holders. The real filter is whether a plant gets built and whether its owner outsources O&M, so the risk-adjusted SAM is the figure to quote. The SOM is a scenario of winning one contract, not a forecast share. It is revenue, not profit. At a 10–15% operator margin, one contract earns CDEG about $2.7–5.9m a year (6.6). Nationally, about 15 WtE projects totalling ~4 Mt/yr are in development [37]. At the same $/t, that is about $350m a year of O&M if all were built. It is upside outside Victoria, not the base case.

The meter market is harder to reach. Melbourne's two largest programs, South East Water (about 0.85–1m meters) and Yarra Valley Water (about 0.9m), are already contracted [65, 66]. Their 1.75–1.9m meters would be worth A$263–285m at a device price ceiling of about A$150 per meter, the price implied by Huizhong's NSW contract for 191,000 meters at up to A$29m [67]. What remains addressable is Greater Western Water, whose joint tender closed in November 2025, and the regional corporations [68]; their connection counts come from the BoM National Performance Report and are still to be added [69]. As a first target, one or two regional corporations taking 50,000 meters in 2028–33 would be worth about A$7.5m [Assumption].

3.3 Technical fit: waste composition and sorting

Victoria's residual waste has almost twice the heating value of Chengdu's 2008 waste and is above Wanxing III's 9.0 MJ/kg design point. It gets hotter as FOGO removes food. It sits inside CDEG's grate window, in the upper half.

How sorting works. Victoria is moving to one statewide four-bin standard: general waste (red), mixed recycling (yellow), FOGO (lime) and glass (purple). The state sets the standard and the levy, and each council runs its own collection and disposal contracts [25, 26]. About 60% of councils offered FOGO in

2026 [20]. Before FOGO, food made up 40–45% of red bins in Knox audits [70] [Industry]. Only the red bin and commercial residual can go to WtE, and both are subject to the permitted-waste rules in Table 2 [10]. CDEG removes metals and non-combustibles but runs no general-waste sorting plant [5]. A local partner should run the sorting step (6.3).

Table 6. Residual waste: Chengdu vs Australia

Measure	Chengdu MSW (2008)	Australian residual, no FOGO (Bromelton, Qld, 2023)	Victorian MSW (Maryvale planning, 2021)
Food / organics	50.3% organics	26.2% food + 12.0% garden	35% food + 7% garden
Plastics	n/a	14.0%	15%
Paper and card	n/a	17.6%	10%
Moisture	48.7%	38.5%	n/a
Lower heating value (LHV)	5.6 MJ/kg	10.6 MJ/kg design (8.8–14.6 range)	n/a

Sources: [71, 72, 73]. Chengdu data predate China's mandatory sorting, so current Chengdu waste is probably drier; Wanxing III is designed for 9.0 MJ/kg [18].

Our estimate. We assigned typical heating values to each material and applied them to the Victorian composition. This gives about 10.2 MJ/kg for red-bin waste without FOGO (fractions and heating values in Appendix E). The same method reproduces Bromelton's 10.6 MJ/kg design point. If FOGO captures 25%, 50% or 80% of the food and garden waste (80% is an upper bound), LHV rises to 10.9, 11.8 or 13.2 MJ/kg, and the tonnes left fall to 90%, 79% or 66% (Figure 4) [Inferred]. Model 2 refines these inputs.

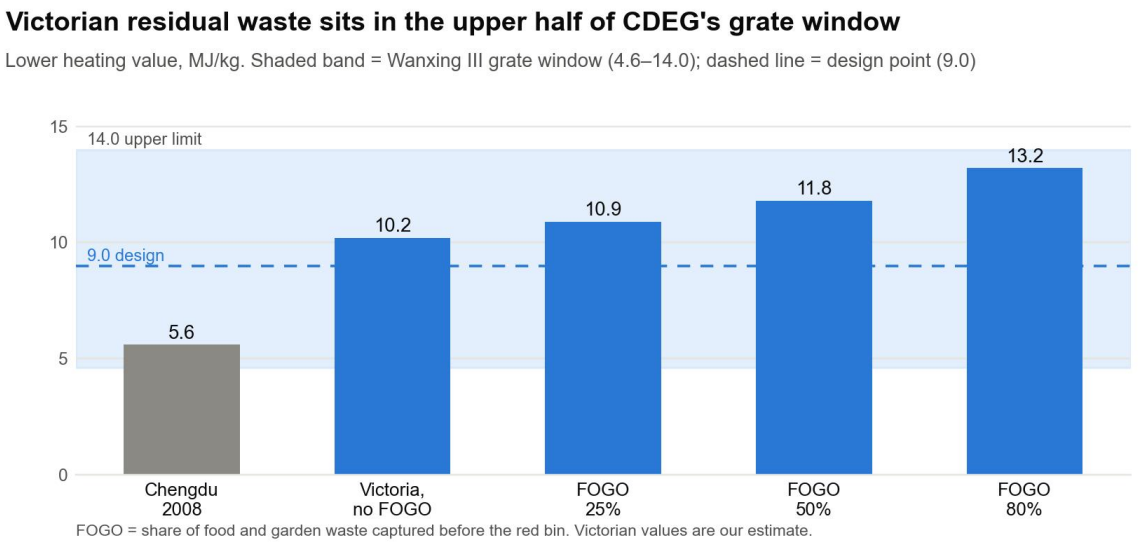


Figure 4. Victorian residual waste against CDEG's grate window

Sources: [18, 73, 72, 71]; Victorian values are our estimate [Inferred].

What it means for CDEG.

- **Fewer tonnes per line.** A grate is limited by heat input. A line rated at 500 t/d for 9 MJ/kg waste takes only about 375–450 t/d at 10–12 MJ/kg [Inferred]. A new Victorian plant would be rated for its own heating value, so this mainly matters when quoting CDEG's 500 t/d minimum: in Victorian tonnes, CDEG's smallest line is larger in heat terms than it looks.
- **A wide operating range.** At 80% FOGO capture, the waste approaches the 14 MJ/kg ceiling. Victorian plants will be designed for a higher point, as Bromelton's 8.8–14.6 MJ/kg range shows [72]. CDEG's experience is running varied waste on a grate, not a specific heating value.
- **More plastic means more acid gas.** A higher plastic share raises chlorine load. That matters most where real-time emissions reporting is coming (2.3).

- **Shrinking feedstock.** FOGO and the 80% diversion target cut residual tonnes. That is a utilisation risk for owners, and it pushes them towards operators with high availability (3.4, 6.7).

3.4 Economics: who pays

A Melbourne WtE plant beats landfill only if it runs at high utilisation (the break-even point is about 72%) and costs no more than about $1,850 per tonne of capacity to build. Utilisation is the lever an operator controls, and it is where CDEG adds value.

This section asks whether build costs, gate fees, power sales and financing terms support CDEG's entry. It uses the only two Australian plants to reach construction, a simple cost model for a plant of two 500 t/d lines, and a scorecard comparing landfill with WtE.

3.4.1 Evidence from Australian plants

Table 7. Australian WtE plants: cost, financing and contracts

	Kwinana (WA)	East Rockingham (WA)
Capital cost	$668m at financial close; about $1.1bn final [74]	$511m
Design capacity	400,000 t/yr	300,000 t/yr plus 30,000 t biosolids
Cost per tonne of yearly capacity	~$1,670 at budget; ~$2,750 final [Inferred]	~$1,550 to $1,700 [Inferred]
Financing	$275m equity (Macquarie, DIF); $400m debt incl. CEFC	Bank debt plus CEFC and ARENA
Government support	$23m ARENA grant; up to $90m CEFC debt	CEFC and ARENA
Waste supply	20-year deals with Rivers Regional Council and City of Kwinana; 5-year Veolia deal	First Australian use of waste-arising contracts
Power offtake	WALGA preferred supplier of baseload renewable power to councils	Part supply to Talison Lithium's Greenbushes site

Sources: [75, 76, 77, 78, 74, 79]; Kwinana final cost [74]. Per-tonne figures are our own division of published totals [Inferred].

Two lessons follow. Budget build costs in Australia run at about $1,550 to $1,700 per tonne of yearly capacity, and Kwinana's overrun shows the final cost can be much higher. Delivery risk is also high: Kwinana took more than six years of work to reach its official opening, and East Rockingham went into receivership [80, 74].

A Columbia University study found Chinese plants cost about US$250 per tonne of yearly capacity, against about US$840 in the United States, and Chinese-involved projects abroad, such as in Vietnam and Ethiopia, about US$350 [81]. The data is from 2018, and costs rise further in Western markets because of labour, regulation and distance from suppliers. We therefore test a 25% capital saving from CDEG involvement, not a China-level cost [Assumption]. CDEG buys its equipment [5], so any saving would come through its Chinese supply chain and EPC partners rather than its own technology [Inferred].

3.4.2 Cost model

The model asks what gate fee a plant needs to cover all its costs, including a return on capital. If that fee is below the landfill price, councils and waste companies have a reason to choose WtE. The inputs come from the team cost model [64], and Appendix E gives the formulas.

Table 8. Cost model inputs

Input	Value	Basis
Plant size	Two 500 t/d lines; 365,000 t/yr design	CDEG minimum line size [5]
Utilisation	Low 70%, base 85%, high 95%	[Assumption]; grate plants usually run

Input	Value	Basis
		85 to 92%
Capital cost	$1,600 per t of yearly capacity; $584m	Kwinana and East Rockingham budget benchmark (Table 7)
CDEG case capital cost	25% lower; $438m	[Assumption] from [81]
Cost of capital and life	8% over 25 years (capital charge 9.37% a year)	[Assumption]
Fixed O&M	2.5% of capital a year	[Assumption]
Variable cost	$40/t (reagents, ash disposal, water)	[Assumption]
Electricity exported	0.6 MWh/t at $90/MWh	[Assumption], in line with [37]
LGCs	50% biogenic share; $15 per LGC	[Assumption]; only the biomass part earns LGCs [82]

Source: team assumptions unless stated.

Table 9. Cost model results

Scenario	Tonnes/yr	Total cost per t	Energy and LGC income per t	Break-even gate fee ($/t)	vs landfill ~$247/t
AU cost, low 70%	255,500	311	58	253	Not viable
AU cost, base 85%	310,250	263	58	205	Viable
AU cost, high 95%	346,750	240	58	181	Viable
CDEG cost, low 70%	255,500	243	58	185	Viable
CDEG cost, base 85%	310,250	208	58	149	Strong
CDEG cost, high 95%	346,750	190	58	131	Strong
Maryvale capex (~$1,850/t), base 85%	310,250	298	58	240	Marginal
Power at $60/MWh, base 85%	310,250	263	41	223	Viable
$20/t WtE levy, base 85%	310,250	263 + 20 levy	58	225	Viable
Kwinana final cost (~$2,750/t), base 85%	310,250	424	58	366	Not viable
Kwinana final cost less 25%, base 85%	310,250	328	58	270	Not viable

Source: team model [Inferred]; landfill price from 2.1. The Kwinana rows use its reported final cost of about $1.1bn [74] [To verify]; the Maryvale, power and levy rows come from the same model (Appendix E).

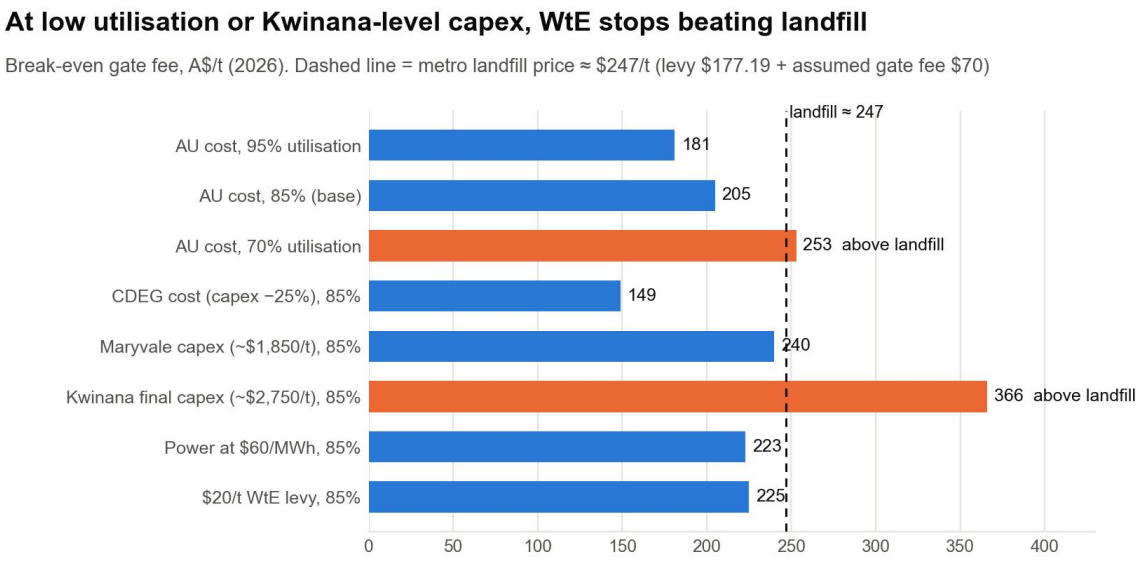


Figure 5. Break-even gate fee by scenario against the landfill price

Source: Model 1 (Appendix E), built on the team cost model [64]; capex benchmarks [37, 74]; levy [2].

What the model shows.

- **Utilisation is the main lever.** Most costs are fixed, so every tonne below capacity raises the cost per tonne. Moving from 85% to 70% adds about $48/t and makes the plant unviable, which is why developers insist on long-term waste supply. Break-even equals the landfill price at about 72% utilisation, and between 85% and 95% each percentage point is worth about $2–3/t [Inferred]. A real levy rise of about 3% would bring even the 70% case level with landfill, so the levy path matters as much as the plant.
- **Gate fees provide most of the revenue.** Energy and LGC income covers only about $58/t, so 75 to 80% of revenue must come from gate fees. This is higher than Harford's 60% estimate in 2.2 because our model counts no metals or other product sales, which Harford puts at about $53/t [37].
- **Capital cost is the next biggest factor.** A 25% saving cuts the base-case break-even by about $56/t, to about $149/t. At Maryvale's ~$1,850/t, break-even is level with landfill. At Kwinana's reported final cost, about $1.1bn for 400,000 t or about $2,750/t against about $1,670/t at budget, the plant stays above the landfill price even with a 25% saving [74] [To verify]. CDEG buys its equipment [5], so this lever sits mainly with the technology vendor, the EPC contractor and CDEG's Chinese supply chain.
- **Policy can erase the margin.** Each rise in the landfill levy widens WtE's margin, while a $20/t WtE levy or power at $60/MWh takes about $20/t off the $42/t headroom (3.5). Victoria's own modelling uses $80.11/MWh [36], and wholesale prices fell sharply in 2026 [22]. Two shocks together would leave almost nothing (6.7).

What it means for CDEG as an operator. CDEG does not sell capex savings. It sells availability and running cost:

- Lifting utilisation from 85% to 90% cuts the owner's break-even by about $12/t.
- A 1% saving on operating cost is worth about $0.27m a year to the owner, or $2.9m in present value over 25 years (Appendix E).

The O&M contract itself is worth about $27m a year for a two-line stage, with CDEG's margin inside that figure. That is the number used in 3.2 and 6.6. In a pitch, these are the figures that show what an experienced operator is worth to an owner whose plant only works at high utilisation.

Limits. This is a simple comparison model, not a bankable forecast. It excludes metals and other product sales, which Harford puts at about $53/t [37]: counting all of them would cut break-even to about $152/t, and a conservative metals-only figure (about $7/t) gives $198/t. It also leaves out tax, inflation, construction delay, a front-end sorting plant, levy on landfilled ash and extra EU-standard gas cleaning beyond the benchmark, which all raise costs. Together these items could move break-even by ±$50/t, which is more than the headroom, so the model shows which levers matter rather than a bankable price.

The inputs should be replaced with CDEG's own capital and operating costs when they arrive (Appendix D).

3.4.3 Revenue, offtake and financing

Gate fees. Kwinana's contracted gate fee is about $120/t, against a much lower WA landfill levy [83]. Wholesale power is cheap and volatile, so the gate fee carries the project [37].

- **Waste supply.** Supply contracts are the core of the business. Kwinana has 20-year council deals, and East Rockingham uses waste-arising contracts that let councils keep pursuing reduction targets [75, 78]. In Victoria, nine south-east councils signed a 25-year agreement for Maryvale [35]. The inquiry warns that minimum guaranteed supply can discourage recycling (Finding 21) and recommends guidance on flexible council contracts (Recommendation 12) [6], so waste-arising contracts may become the expected model [Inferred]. In October 2025 Cleanaway agreed to supply waste feedstock to Kwinana [84], which shows that large waste companies supply WtE plants as well as develop them.
- **Power offtake.** Kwinana became WALGA's preferred supplier of baseload renewable power to councils, and East Rockingham signed with Talison Lithium [78, 74]. In Victoria, buyers could be councils, large industrial users or, at Maryvale, the paper mill itself [Inferred].
- **LGCs.** Only the biomass part of municipal waste earns LGCs, and plastics do not count [82]. At a 50% biogenic share, LGCs add only about $4.50/t.
- **Financing.** Kwinana used about 40% equity and 60% debt, with the CEFC as a key lender, and was later refinanced with a sustainability-linked loan that repaid the CEFC [76, 77]. Lenders will want long-term waste and power contracts, proven technology and an experienced operator. CDEG's record of running about 12,000 t/d of grate capacity in Chengdu addresses the last point.

3.4.4 Landfill versus incineration scorecard

Scores run from 1 (poor) to 5 (good) and are our judgement from the evidence above [Inferred].

Table 10. Landfill versus WtE scorecard

Criterion	Landfill	Score	WtE	Score
Cost per tonne (metro)	~$247 and rising with levy	2	~$150 to $205 in base case	4
Price certainty	Levy set by government, rises often	2	Fixed in long-term contract	4
Greenhouse gases	Methane from organics	2	CO2 from plastics; better only while landfill takes organics (Findings 34, 35)	3
Residues	Waste stays in the ground	2	20 to 25% ash; fly ash hazardous (Findings 4 to 7)	3
Energy recovery	Small, from landfill gas	1	About 0.6 MWh per tonne	4
Land use	Large and growing	1	Compact site	4
Community acceptance	Accepted but disliked	3	Strong opposition, e.g. Sunbury (4.4)	2
Flexibility	Takes any volume	5	Needs steady supply; lock-in risk (Finding 21)	2
Policy risk	Levy keeps rising	3	Possible WtE levy, election, cap limits	2
Total (out of 45)		21		28

Source: team judgement from Sections 2 to 4; findings are from the inquiry report [6].

WtE scores better on cost, energy and land, but worse on flexibility, community acceptance and policy risk. CDEG's offer needs to address these weak points: flexible supply contracts, EU-level emissions data and transparent community engagement (6.2).

3.5 Government support and policy risk

Federal agencies (ARENA and the CEFC) helped finance the two WA plants, but Victoria's main grant programs exclude thermal WtE.

Table 11. Support programs

Program	Level	Supports thermal WtE?	Relevance to CDEG
ARENA grants	Federal	Yes in the past ($23m to Kwinana)	Possible for innovative parts, e.g. CCUS
CEFC	Federal	Yes (up to $90m debt to Kwinana)	Debt finance for a partner's project
Recycling Modernisation Fund	State	No for energy-focused projects	Material recovery or chemical recycling only
Bioenergy Fund ($10m)	State	No if a cap licence is needed	Could fit food waste and anaerobic digestion
Sustainability Fund	State	No; proposal to open it to WtE voted down 3 to 4	None for now
LGCs	Federal	Partly, biogenic share only	Small income

Sources: [6, 76, 82, 7, 85].

Four policy risks could change margins:

- **A possible WtE levy.** Finding 23 and Recommendation 13 ask the government to review whether economic incentives still favour recycling over thermal treatment [6].
- **A federal incinerator levy.** The Greens proposed a $20/t levy on municipal waste sent to incinerators at the 2025 federal election [86].
- **The state election.** The Victorian election on 28 November 2026 could change WtE policy.
- **CDEG's own history.** Its Pakistan and Uzbekistan projects stopped for policy reasons (2.2), so policy risk should be weighted heavily.

In the base case, a $20/t WtE levy would push the Australian-cost break-even from about $205/t to about $225/t, still below landfill at about $247/t. With CDEG's cost saving the margin stays comfortable [Inferred].

3.6 SWOT and TOWS



Figure 6. SWOT: CDEG in Victoria

Four strategies follow from the SWOT. Matching strengths to opportunities, CDEG should offer O&M to licence holders, backed by verified EU-format emissions and its co-processing record. Against the threats, it should lead with public live-emissions data and keep its branding partner-led. To cover its weaknesses, it should join a consortium with a European or Japanese technology supplier or a local EPC, borrowing an OECD record. Where weaknesses meet threats, it should keep its first moves asset-light: product sales through a distributor and joint R&D.

4.0 Customer Insights

We did no interviews, so each persona rests on what that customer has published: inquiry submissions and hearings, ASX announcements, tender notices and regulator records.

4.1 Persona: project director at a cap-licence holder

Four holders matter. Cleanaway (760,000 t/yr) is re-filing its EPA licence for the larger Wollert plant, runs an open technology procurement and may operate the plant itself; Whittlesea Council opposes the project [1, 52]. HiQ (750,000 t/yr) has chosen Kanadevia Inova for its Bulla plant, still has its applications to lodge and targets full operation around late 2030 [87, 88]. Melbourne Water (450,000 t/yr) is seeking a biosolids solution and called for EOIs in April 2025, with a 70,000 t minimum [89, 54]. REL (280,000 t/yr) plans gasification at Laverton North; it holds both licences and says it is building, but has named no supplier, start date or operator [1, 55]. None has named an operator (Appendix A).

The project director at each of them must reach a final investment decision, satisfy the lenders' technical adviser, meet EPA conditions and win over the community. Their fears are cost overruns, delay and a plant that cannot run at full load. Kwinana, Australia's first large plant, was budgeted at $680m and cost more than $1bn [37]. They buy on an OECD track record, availability guarantees, a local workforce with a good safety record, and price.

4.2 How WtE buyers decide

The owner chooses the technology, the EPC contractor and the operator, and buys on price, guarantees and bankability. Behind it sits the lenders' technical adviser, who reviews the operator's track record, availability, liquidated damages and parent guarantee; CEFC alone lent $400m to Kwinana [37]. EPA licenses the plant and sets its emission and monitoring conditions. Councils supply the waste on 25-year

contracts and compare the gate fee with the cost of landfill [35]. The community shapes planning decisions and politics through its concerns about health, traffic and transparency (4.4).

4.3 Persona: digital lead at a regional water corporation

This buyer needs leak detection, accurate billing and data that flows into its platform, often Iota's Lentic; Barwon Water chose Iota [90]. Its budget is set in the ESC price review: submissions for 2028–33 are due on 13 September 2027 and the decision comes in June 2028 [27, 13]. It runs pilots through Intelligent Water Networks, a collaboration of water corporations, VicWater and DEECA that lists digital metering among its programs [91]. A supplier must hold NMI pattern approval under NMI R 49, plus AS/NZS 4020 and WaterMark compliance [92, 93].

Public discussion of digital meters is small and practical. Across 177 forum posts, 42% are about installation, 39% about leaks and usage data and 14% about bills; privacy appears in 6% [94]. For a meter supplier, the barrier is procurement and certification, not public acceptance.

4.4 Public sentiment

The WtE plant in Victoria not only requires a license but also community acceptance, which has an impact on the subsequent actions of the corresponding operators. With no interviews at this stage, submissions to that inquiry are the public record of what residents and industry think. This section answers two questions: what residents fear most, and whether Chinese involvement adds to it.

We coded the 565 readable submissions to the WtE inquiry: 537 from individuals and community groups and 16 from industry [8]. Health and air pollution is the leading community concern (86%), followed by local amenity and siting (73%), recycling and the waste hierarchy (39%), consultation, trust and regulation (27%), and cost and contracts (23%). Industry argues in other terms: recycling and the waste hierarchy (94%) and the energy and landfill-diversion benefits of WtE (81%).

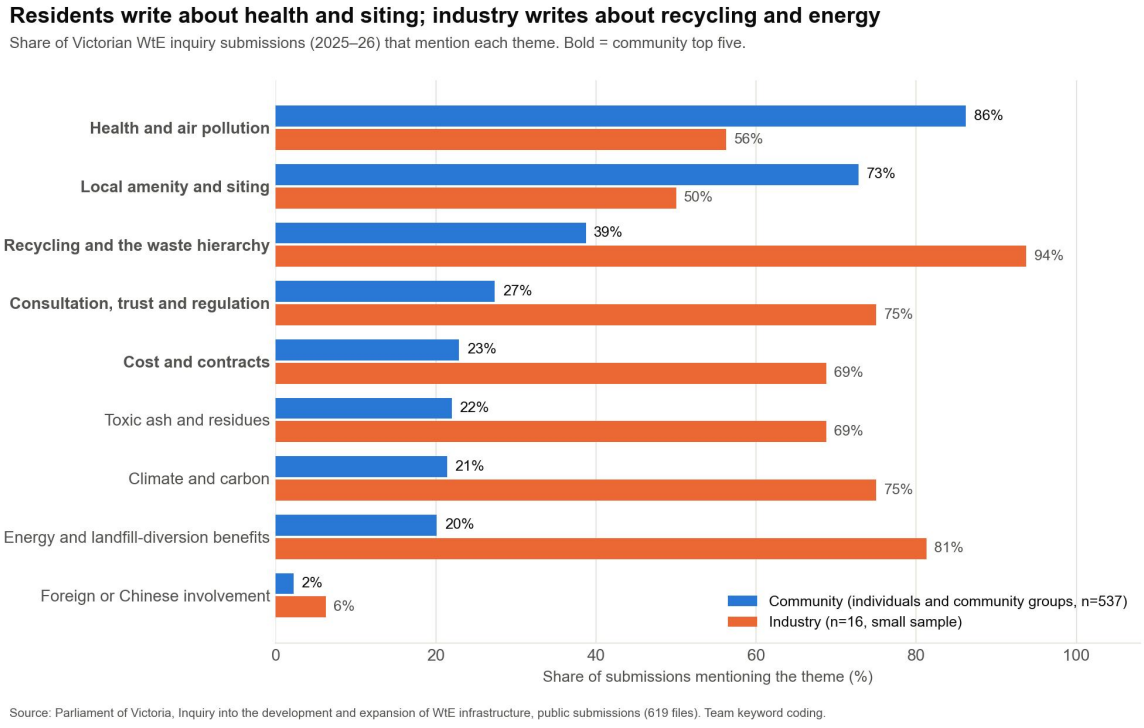


Figure 7. Themes in WtE inquiry submissions, community vs industry

Opposition is local. 56% of community submissions name Sunbury or Bulla (HiQ) and 14% name Wollert, Epping or Donnybrook (Cleanaway). In contrast, Chinese involvement is rarely mentioned. Among 537 community opinions, only 4 expressed concerns about foreign ownership or investment, 3 used Chinese factories as evidence of emissions, and 2 presented China as a positive example.

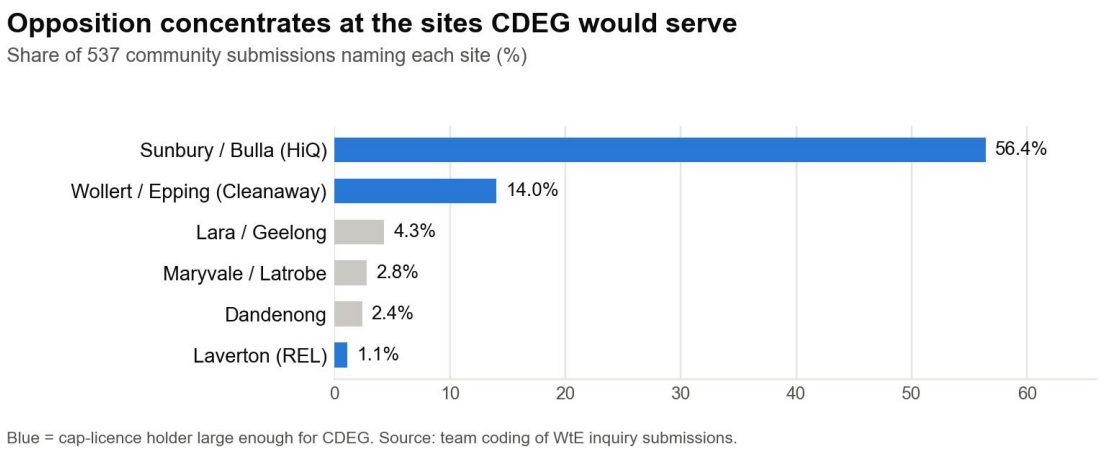


Figure 8. WtE sites named in community submissions

Therefore, an O&M partner at HiQ or Cleanaway will inherit this opposition. CDEG's offer should lead with verified emissions and live public data, then truck and siting management. However, this data analysis also has significant limitations. This is not a general opinion poll, because the submitters are self-selected and the majority are opponents. These figures only indicate concern.

5.0 Competitor Analysis

5.1 Value-chain role map



Figure 9. Victorian WtE value chain and the roles open to CDEG

Four of the six roles are closed or a weak fit. The role of operators is open, because most license holders have not named operators. Where one has been named, it sits inside the ownership group: Veolia is a partner in the Maryvale consortium [95], and ACCIONA operates its own Kwinana plant [80]. So we inferred that EPC is open only inside a consortium.The full license holder matrix is shown in Appendix A [1]

5.2 Australian incumbents: business and current status

Four groups shape the operator market: one national waste company, two European operators and the technology vendors.

- **Cleanaway, the national leader**: In FY26 it earned net revenue of $3.74bn (up 13.1%) and underlying EBIT of $470m (up 14.2%, a 12.6% margin) from more than 350 locations, 6,200 vehicles and 9,700 staff [96]. Its Solid Waste Services, which covers collection, landfills and the container deposit scheme, made $2.51bn of revenue and $405m of EBIT. It holds the 760,000 t/yr cap licence

for Wollert [1], but withdrew its EPA development licence application in September 2025 to re-file for the larger plant [23]. In January 2026 it also signed a joint development agreement for the 700,000 t/yr Parkes plant in NSW with Tadweer and Tribe [97]. On 18 October 2026 it agreed to be bought by EQT Infrastructure for $3.095 a share, an enterprise value of about $9.3bn, with completion expected by March 2027 [24]. EQT owned EEW, Germany's largest WtE operator, until it sold it to Beijing Enterprises in 2016 [98]. Wollert's technology and operator choices may now wait for the new owner **[Inferred].**

- **Veolia, the operator with the widest reach**: The French group's Australian arm runs water, waste and energy services and operates 68 water, wastewater and recycled-water plants. In November 2025 it won $700m of water contracts, including 15 years at six Central Highlands Water treatment plants in Victoria [99]. It is the operating partner in the Maryvale consortium with Opal and Masdar Tribe, to which more than a dozen councils have committed waste, nine of them on a 25-year agreement, but we found no announcement of financial close [60, 35, 95]. It ran Kwinana's O&M until early 2025, and its Woodlawn plant in NSW (380,000 t/yr) is still under assessment, with the Independent Planning Commission to decide [37, 100].
- **ACCIONA, the only WtE owner-operator**: It owns and operates Kwinana, Australia's first large plant, which cost more than $1bn against a $680m budget [80, 37]. At East Rockingham it is EPC contractor, creditor and 10% shareholder [80]. Its bid for that plant drew ACCC competition concerns, the review was withdrawn on 13 October 2025, and the plant was still with receivers at last report [101, 80].
- **Other waste groups**: Remondis withdrew its $400m Swanbank EfW application in Queensland in 2022 after the Coordinator-General found that a regional planning instrument made it unviable [102, 103]. Visy holds an existing-operator licence at Coolaroo [1].
- **Technology vendors**: Kanadevia Inova (HiQ, Parkes), Keppel (Kwinana) and Babcock & Wilcox (Maryvale) can bundle O&M with their grates [87, 37, 95].

Overall, the operating market is thin. One large plant runs, one is in receivership and Victoria's plants start from 2029. The two experienced operators sit inside consortia, and Cleanaway is changing hands. CDEG's opening is an O&M or technical-services role for a licence holder without an operator (HiQ, Melbourne Water, REL), or a partner role under Cleanaway's new owner **[Inferred]**.

5.3 Chinese SOEs in Australia: case learning

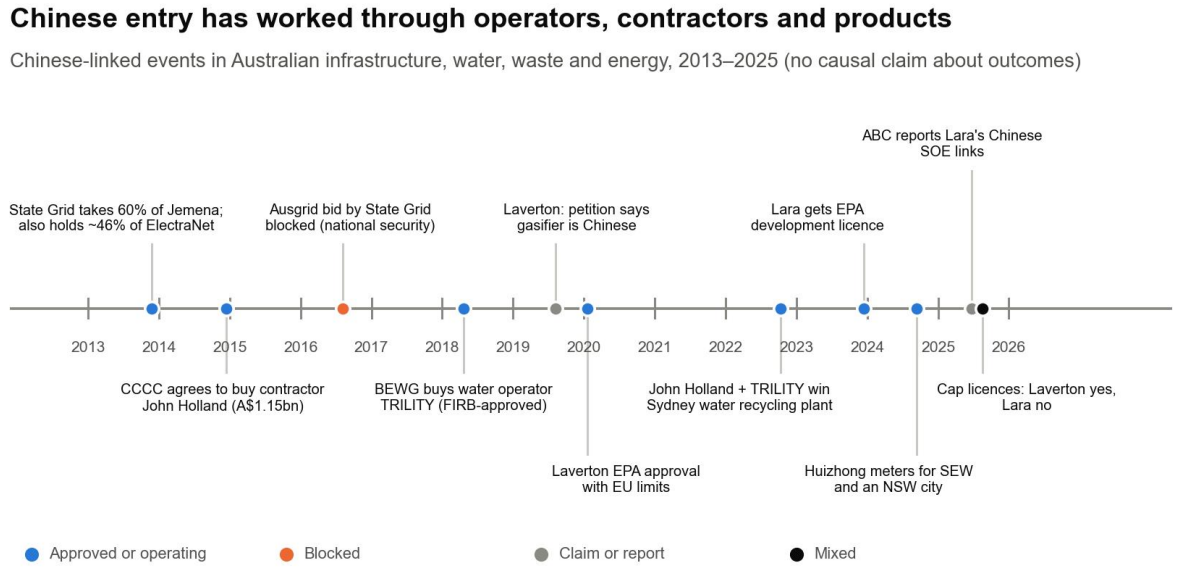


Figure 10. Chinese-linked events in Australian infrastructure, water, waste and energy

Chinese state-owned firms have entered Australian infrastructure in three ways. The closer a case is to a service business run by Australian managers, the better it has gone.

- **BEWG and TRILITY, water operations (reference value: high)**: Beijing Enterprises Water, a Beijing municipal SOE, bought 100% of TRILITY with FIRB approval in 2018 [104]. TRILITY kept its

Australian name and team. It operates and maintains the Adelaide Desalination Plant through the AdelaideAqua joint venture until 2033 [105]. In 2022 it joined John Holland, GHD and Jacobs to win Sydney Water's Upper South Creek recycling plant, which it will operate with John Holland for up to 10 years [106]. For CDEG, also a city-owned group, this is the closest model: run services under Australian management and win contracts inside a local consortium.

- **CCCC and John Holland, construction (reference value: medium)**: China Communications Construction agreed in December 2014 to buy John Holland from Leighton for $1.15bn [107]. John Holland still wins large public contracts under its own brand, but it lost $117m in its latest reported year, up from $34m, after write-downs on Melbourne's West Gate Tunnel and stage 3 of the Gold Coast light rail [108]. Fixed-price construction can erase margins, which supports CDEG's preference for O&M over EPC risk.
- **State Grid, energy networks (reference value: low for control)**: State Grid, a central SOE, holds 60% of Jemena beside Singapore Power's 40% [109]. Jemena owns gas networks and Jemena Electricity Networks in Victoria [110]. It also holds about 46% of ElectraNet, South Australia's transmission network, alongside Australian investors [111]. However, its 2016 bid for a controlling lease of NSW's Ausgrid was blocked on national-security grounds [112]. The dividing line is timing and control rather than stake size: these stakes were approved in 2012–13 and each sits beside a non-Chinese co-owner, while the 2016 bid for control of Ausgrid was refused. The owners of critical electricity, gas and water assets must now also run a risk-management program under the SOCI Act [49], and these duties flow down to contractors (6.3). Therefore, an O&M contract is open to CDEG, but the equity or control in such assets would face the hardest review.
- **WtE precedents (reference value: medium)**: Lara is the only Victorian project reported to involve a major Chinese WtE firm: company documents named Everbright as technology partner and a CEEC institute as designer [113]. Its small developer holds an EPA development licence but missed out on a cap licence, still has no planning permit and had no confirmed financing [1, 113, 114]. Laverton North, whose gasifier supplier is unnamed, holds both licences and says it is building, but has given no start date or operator [1, 55]. The lesson is to partner with a funded licence holder, not a small developer.

In meters, Huizhong supplies South East Water through the distributor Avnet [115, 67], and Sanchuan and Ningbo Water Meter already hold NMI approvals or contracts [12], so CDEG would arrive late.

Overall, CDEG can enter as a service operator under Australian management, inside a consortium with Australian or other non-Chinese partners, and keep any equity small. This is the O&M-led, asset-light entry in 6.3. TRILITY and John Holland, both Chinese-owned local firms, are worth approaching as partners.

5.4 Positioning and Five Forces

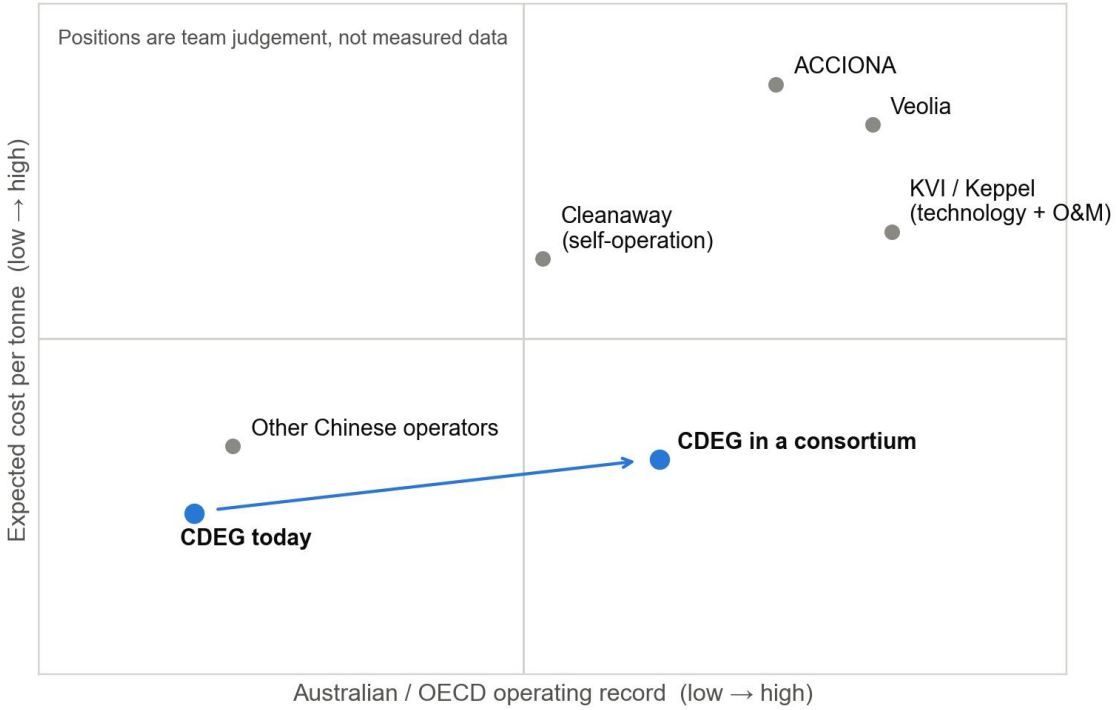


Figure 11. Positioning of WtE operators in Australia (team judgement)

CDEG's low-cost position is a hypothesis until Model 1 and CDEG's cost data confirm it. Its gap is the Australian record, which a consortium partner can supply.

The Five Forces point the same way. Buyer power is high, because four large buyers sign long-term contracts. Supplier power is medium to high, because technology vendors can bundle O&M with their grates. Rivalry is medium: there are few operators, and the plants are years away. Substitutes are medium, as landfill remains until the mid-2030s alongside digestion and recycling. Entry barriers are high: the cap, the need for a track record and FIRB. CDEG should therefore enter with a technology partner or EPC, not alone.

6.0 Go-to-Market Plan

6.1 Priority segments

Each option first passes three gates: a defined product, a compliance path and evidence of demand. It is then rated on five weighted criteria: demand and reach 25%, technical fit 25%, economics 25%, partners 15% and pilot speed 10% [116]. Where CDEG's data is missing we rate qualitatively and leave a question mark, rather than giving a midpoint score.

Table 12. Screening of entry options (H = high, M = medium, L = low, ? = data pending; Fit = technical fit)

Option	Gates	Demand	Fit	Econ.	Partners	Speed	Result
WtE O&M and technical services	Pass	M	H	M	M	L	Priority 1 (strategic)
Biosolids and sludge with Melbourne Water	Pass	M	H	?	M	L–M	Priority 2
Water digital products, regional corporations	Pass	H	M	M	M	M	Priority 3 (near term)
Food-waste and organics equipment	Pass	H	L–M	?	M	M	Secondary

Option	Gates	Demand	Fit	Econ.	Partners	Speed	Result
Joint R&D (CCUS, sludge, AI)	Pass	L–M	M	H	H	H	Enabler
Technology licensing or introduction	?	L–M	?	?	L	L	Later track
Co-investment or JV	Fail now	—	—	—	—	—	Later track
Coal-site conversion (Latrobe)	?	M	M	?	L–M	L	Later track
Water and wastewater plant O&M	?	L	H	?	L	L	Watch

Co-investment fails the near-term gates on FIRB, China's outbound approval and the allocated cap. A robustness test of the weights follows once CDEG's data fills the question marks.

The priority segments are (1) the four large licence holders, for O&M and technical services, with Cleanaway, HiQ and Melbourne Water ahead of REL, which uses gasification; (2) Melbourne Water, for biosolids co-processing; and (3) regional water corporations, for digital meters and drainage IoT. Joint R&D runs alongside all three.

6.2 Value proposition and positioning

The canvas turns Section 4 and 5 into CDEG's offer for the first priority segment: the four licence holders large enough for a 500 t/d line [1, 5]. The right side shows what a licence holder needs; the left side shows what CDEG would sell. The fit rests on three matches. An operator with a long record, a parent guarantee and an Australian-managed team answers the fear of overruns and of an unproven operator, at a time when the operator market is thin. A live public emissions dashboard answers local opposition. Biosolids co-processing adds feedstock for Melbourne Water. The weak link is cost per tonne, which stays a hypothesis until CDEG's cost data confirm it.

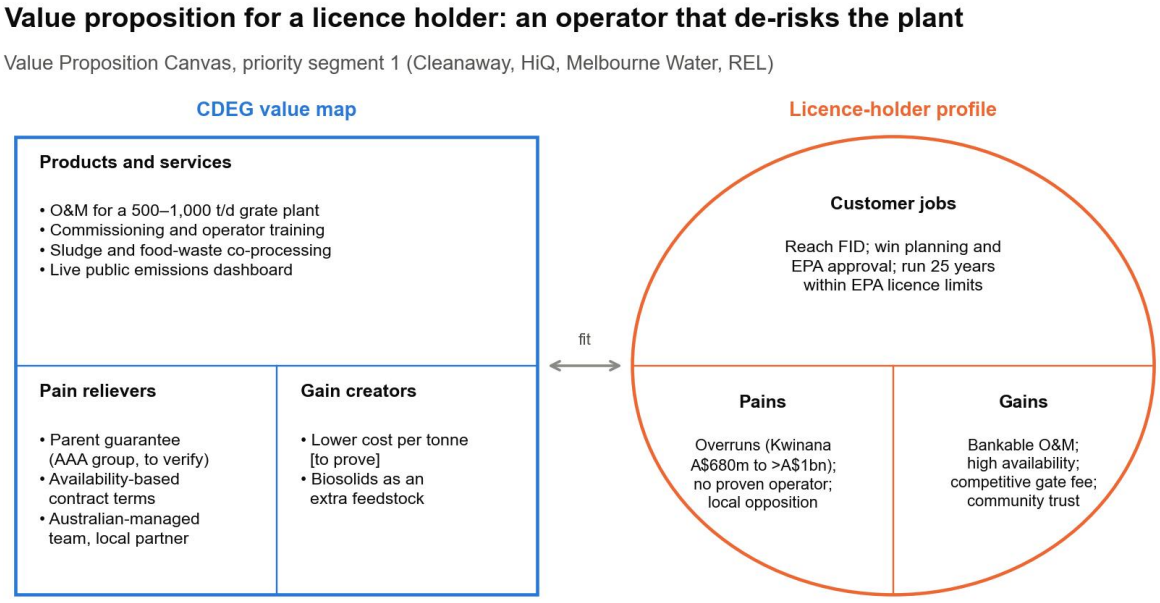


Figure 12. Value Proposition Canvas for licence holders (priority segment 1)

For Melbourne Water, which holds a 450,000 t/yr cap licence but has not named an operator, the job is processing biosolids at a lower disposal cost. CDEG's answer is sludge co-processing designed at 800 t/d at Wanxing III [5, 15], with water and waste know-how in one group. The proof needed is operating and emissions data from its sludge lines.

For regional water corporations, the job is finding leaks and billing accurately at low cost. CDEG's answer is low-cost NB-IoT ultrasonic meters and drainage sensors. Huizhong, Sanchuan and Ningbo already hold

approvals or contracts, so price alone will not win. CDEG must bundle leak analytics and local service with the meters **[Inferred]**. The proof needed is NMI approval, an integration test with Iota's Lentic platform and pilot results in time for the 2028 Water Price Review [13].

Positioning statement: for Australian owners of large waste and water assets, CDEG is the operator that has run Asia-scale waste plants and co-processes sludge, offered asset-light under Australian management with a local partner, and backed by verified, public emissions data.

6.3 Entry pathway



Figure 13. Entry-mode ladder with timing and approvals

CDEG moves up the ladder only where the risks allow. Steps 1 and 2 need no Australian entity: meters are sold through a distributor once NMI approval is in hand, and R&D runs as a research contract. From step 3, CDEG needs a Melbourne subsidiary, which requires FIRB approval, with a median wait of 35 days in early 2026, and China-side outbound filing [14, 48, 50].

Step 4 starts when the licensed plants need operators, from about 2029, with CDEG bidding inside a local consortium. Step 5 stays a minority stake, because control of critical assets faces the hardest review.

6.4 Channels, marketing and sales

CDEG needs four kinds of channel. Consortium partners give it an OECD record: technology suppliers such as KVI and Keppel, local EPCs, and engineering advisers such as GHD, Arup, Jacobs, WSP and Aurecon; it should approach two of them by Q1 2027. Government and program channels open doors: Invest Victoria, Austrade, the City of Melbourne and ACASE, starting in pitch week. Industry events give visibility: All Energy Australia (28–29 October 2026), Waste Expo, WMRR, Ozwater and VicWater.

For meters, Intelligent Water Networks is the pilot channel [91], and a distributor with testing and installation capability, such as Service Stream or NPS, handles NMI testing and after-sales. Universities such as RMIT, funded through ARC Linkage or CRC-P grants, provide joint R&D and credibility. The marketing assets are an English technical dossier (KPIs, EU-format CEMS data, case studies), reference visits to Wanxing III and a public live-emissions dashboard.

6.5 Roadmap and KPIs

CDEG should enter in phases tied to Victoria's own calendar. It moves to the next phase only when a gate is passed, and it stops if any of the stop conditions below is met.

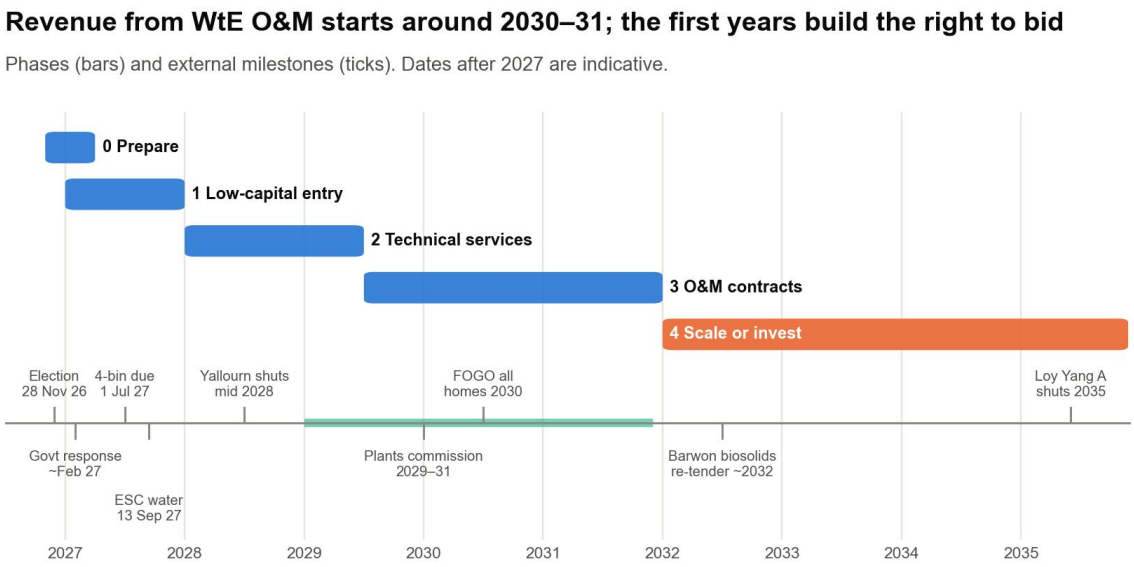


Figure 14. Roadmap: CDEG's phases and gates against Victoria's calendar

Sources: election [51]; government response [6]; four-bin deadline [26]; ESC [27]; Yallourn and Loy Yang A [58, 59]; plants (Appendix A); FOGO [25]; Barwon [22] [To verify].

Table 13. Phases, stage gates and KPIs

Phase	Main actions	Gate to move on	KPIs (target)
0 Prepare Nov 2026–Mar 2027	Verify Wanxing and Longfeng CEMS data in EU format, with a third-party check (Appendix C). Prepare an English technical dossier. Hold partner talks with a local waste company, which brings sorting and supply contracts, and an engineering firm. Watch the election and the inquiry response. Ladder step 1 (6.3).	Emissions data meet EU daily limits. No moratorium after the election.	Lead: verified dossier sent to 4 licence holders and Melbourne Water; 6 partner meetings. Result: 2 MoUs (consortium partner, university)
1 Low-capital entry 2027	FIRB approval and China filings for a Melbourne Pty Ltd (6.3). Biosolids co-processing proposal to Melbourne Water. Join a consortium's response to Cleanaway's technology EOI as the operator, not as a technology owner. Food-waste decontamination pilot. Water: submit one meter for NMI approval (~12 months) [117] and brief regional water corporations before ESC submissions close on 13 Sep 2027 [27]. Ladder steps 1–3 (6.3).	Entity set up. At least one consortium agreement. Shortlisted for one bid.	Lead: 3 bids or proposals submitted; 2 water corporations briefed. Result: FIRB decision ≤ 3 months; 1 pilot running; 1 shortlist; NMI application lodged
2 Technical services 2028–mid 2029	Paid owner's-engineer, operations-readiness or commissioning advice to licence holders during their EPA development-licence and design stage. Build an Australian team under local management. Water: first meter sales after NMI approval, into the 2028 price period. Ladder step 3.	A signed O&M term sheet, or preferred-operator status on one plant.	Lead: 2 paid advisory contracts. Result: services revenue covers local overhead; 8+ local staff; zero safety incidents
3 O&M contracts mid 2029–2031	Mobilise as the plant commissions. Run a public live-emissions dashboard. Report against contract KPIs. Ladder step 4.	Plant meets availability and emissions guarantees for 12 months.	Availability ≥ 90%; 100% of days within licence limits; zero lost-time injuries; contract renewed or extended
4 Scale or invest 2032+	A second O&M contract. Bid the biosolids re-tenders. Only once risks are clear, consider a minority stake, technology	Separate FIRB and NDRC approval for any equity.	2+ contracts; equity decision gated by 6.1 scoring

Phase	Main actions	Gate to move on	KPIs (target)
	introduction or the Latrobe hub (Model 3). Ladder step 5.		

Sources: [5, 14, 24, 48, 50, 6, 54, 52, 117]; targets are the team's proposals [Assumption].

Timing. We show first O&M revenue in 2030–31. Because no licensed plant has reached construction and Kwinana took about seven years [64], a 2032 start is a realistic downside. Phases 1–2 must carry the team until then.

Stop conditions. CDEG should pause if any of these happens:

- Its emissions data cannot be verified against EU daily limits.
- No local consortium partner signs by the end of 2027.
- The new government freezes WtE approvals or imposes a levy large enough to make plants unviable (Figure 5).
- No licence holder will outsource O&M by the end of phase 2.

6.6 Budget and feasibility

Entry costs about $9–10m over three years, including contingency. One O&M contract repays that in about two to four years of operation.

Table 14. Indicative entry budget (A$m)

Item	Year 1 (2027)	Year 2	Year 3	Basis
Melbourne office and staff (3 in Y1, rising to 8)	0.9	1.6	2.2	Country manager, plant engineer, bid lead; local salaries [Assumption]
Legal, FIRB, entity, tax, SOCI compliance	0.3	0.15	0.15	FIRB fee and advisers [To verify]
Emissions verification and English dossier	0.15	0.05	0.05	Third-party CEMS audit [Assumption]
Bids and consortium costs	0.3	0.6	0.6	EOI and tender responses, partner studies [Assumption]
Pilots (food waste; 1,000 meters at ~$150)	0.3	0.2	—	Meter price from [67]; installation [Assumption]
NMI testing and approval (one meter)	0.15	—	—	China test report ~RMB 200,000 (CDEG estimate in team assumptions file) [22]; NMI fees [To verify]
Events, travel, university R&D seed	0.2	0.2	0.2	All Energy, Waste Expo, Ozwater; one research contract
Contingency (15%)	0.35	0.4	0.5	
Total	2.7	3.2	3.7	$9.6m cumulative; $8.3m before contingency

All figures are team estimates [Assumption], to replace with CDEG's own costs and adviser quotes. Not included: mobilising a plant crew before commissioning (paid by the owner under most O&M contracts, To verify), performance bonds and a parent guarantee, which need SAFE registration (6.2).

Table 15. O&M contract P&L: two-line stage (310,000 t/yr at 85%)

Line	A$m/yr	Basis
O&M contract revenue	27.0	$87/t × 310,250 t (Model 1)
Operating cost (staff, maintenance, consumables)	23.0–24.3	Revenue less margin
Operator margin (10–15%)	2.7–4.1	Typical service margin [Assumption]

Line	A$m/yr	Basis
Melbourne Water scale (450,000 t)	3.9–5.9	Same margin, larger plant

Source: Model 1 (Appendix E). $87/t is the owner's operating cost under the contract, with the operator's margin inside it. The fee is treated as cost-plus; availability bonuses and penalties, and liability under performance guarantees, are not included (risk 12).

Feasibility. CDEG's group revenue is about RMB 12.9bn and Xingrong's 2025 net profit was about RMB 2.0bn [3, 15], so a $9–10m entry budget is small for CDEG. At a $2.7–4.1m annual margin, one contract repays it in about 2.3–3.6 years. The real constraint is time: no WtE O&M revenue arrives before about 2030. The phase 1–2 moves (pilots, biosolids, technical services) keep the team funded and visible in the meantime. That is why the plan has two speeds.

6.7 Risks and mitigation

The largest risks are political and social, not technical. CDEG manages them by staying a contractor, publishing its emissions data and entering beside a local partner. Most can be managed through contract terms and the asset-light entry in 6.3.

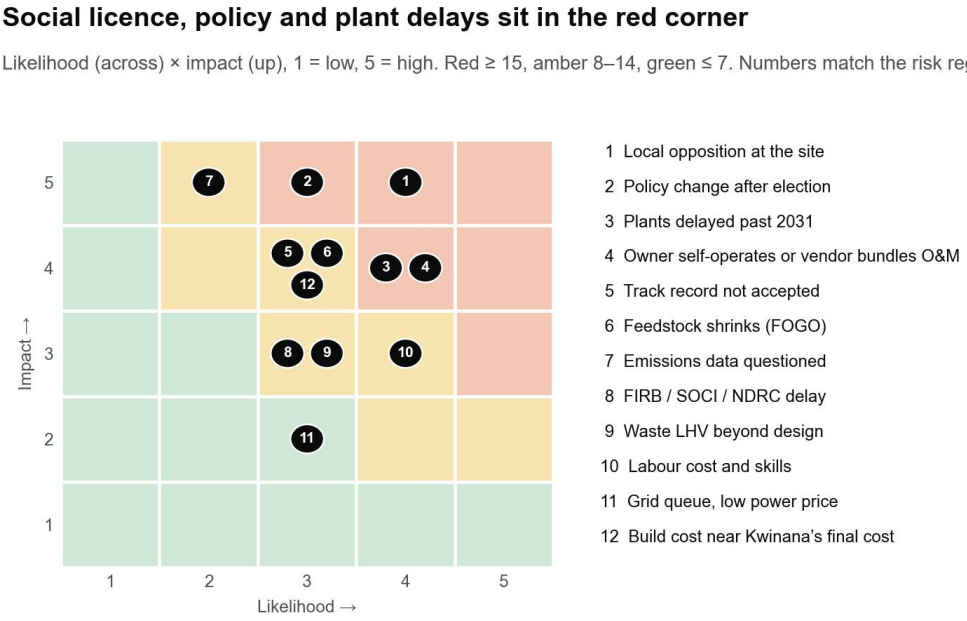


Figure 15. Risk heat map

Team scoring [Inferred]; evidence in Table 16.

Table 16. Risk register with mitigations

#	Risk	Evidence	L × I (1–5)	Mitigation
1	Local opposition blocks or delays the host plant	86% of community submissions raise health; 56% name Sunbury/Bulla (4.4) [8]	4 × 5	Prefer Melbourne Water and Cleanaway before HiQ; live public emissions dashboard; truck and odour plans; local hiring; local partner
2	Policy change after the 28 Nov election or in the inquiry response (WtE levy, project scrapped)	Recs 13, 19, 22 [6]; Opposition pledge on Sunbury (3.1a)	3 × 5	Stay asset-light; gate phases on the response; price-adjustment clauses in O&M contracts
3	Plants slip past 2031, so no O&M revenue	No plant has reached construction; Kwinana took about 7 years and ran over budget [37, 118]	4 × 4	Bridge revenue from biosolids, organics and technical services (6.5); owners need an experienced EPC on a fixed-price contract

#	Risk	Evidence	L × I (1–5)	Mitigation
4	Owners self-operate or vendors bundle O&M	ACCIONA runs Kwinana; Veolia sits in the Maryvale consortium; vendors can bundle O&M (5.1–5.2)	4 × 4	Partner with a vendor rather than compete; offer availability guarantees the owner cannot match
5	Lenders reject CDEG's China-only track record	No plant outside China (1.2); lenders want proven operators [64]	3 × 4	Consortium with an Australian operator; parent guarantee; verified operating KPIs
6	Feedstock shrinks as FOGO and recycling rise	Residual tonnes fall to 66–90% (3.3); inquiry Rec 12 on flexible contracts [6]	3 × 4	Price the fee per available line, not per tonne; size in stages; waste-arising contracts and a commercial waste top-up
7	Public emissions data are questioned, or standards prove stricter than China's	Inquiry doubts proponent-supplied data [6] (Finding 41); raw CEMS data not yet received (Appendix C); EU limits are tighter than GB 18485 (Appendix C)	2 × 5	Request raw CEMS data and a third-party audit before any pitch (phase 0); verified CEMS data in EU format
8	FIRB, SOCI or China outbound approvals are slow	$0 threshold for SOEs; median 35 days in early 2026; new NDRC rules from 1 Dec 2026 [48, 50]	3 × 3	Apply early; start with contracts that need no entity (6.3); O&M or technology licensing over equity
9	Waste LHV runs above the design range	10–13 MJ/kg against a 14 MJ/kg ceiling (3.3)	3 × 3	Design-stage advice; blend with biosolids, as Wanxing co-fires sludge
10	Australian labour cost and skills	Chengdu staffing ratios will not transfer [Inferred]	4 × 3	Local management with a small expatriate core; automation and remote monitoring
11	Grid queue, curtailment, low power prices	Victorian wholesale prices fell ~60% y/y in Q2 2026 [22]	3 × 2	The owner's risk under O&M; keep the fee independent of power price
12	Build cost closer to Kwinana's final cost	Kwinana cost about $1.1bn against $668m at financial close (Table 7) [74]	3 × 4	Fixed-price EPC; CDEG cost data to test the saving
13	Liability under availability and emissions guarantees	Owners and lenders will ask for guarantees (risk 5)	3 × 3	Cap liability at a share of annual fee; parent guarantee sized to the contract
14	Australia–China relations; AUD/RMB exchange rate	Ausgrid refusal (5.3) [112]	2 × 4	Australian management and partner; contracts and costs in AUD

Table 17. Three policy scenarios after the election and the inquiry response

Scenario	What happens	Effect on the plan	Signal to watch
A. Continuity (base)	The cap stands. The government accepts monitoring and transparency recommendations. Plants proceed slowly.	Roadmap as in 6.5; first O&M around 2030	Response accepts Recs 19 and 22 without a moratorium
B. Tightened	A WtE levy (e.g. $20/t), stricter buffers, mandatory real-time reporting.	Headroom roughly halves (3.4). Fewer plants are built. Operators with verified data gain share	Levy announced in the 2027–28 Budget; buffer rules issued
C. Reset	A new government scraps contested projects (such as Bulla) or pauses the scheme.	WtE track deferred. CDEG keeps the biosolids, organics and water tracks	Election result; Bulla decision; licences cancelled or reallocated

Who can block a project: EPA, the Planning Minister and councils, community groups (including VCAT appeals), FIRB and the Treasurer, and lenders. Risks 13–14 are not plotted on the heat map.

7.0 Recommendations and Next Steps

Enter asset-light in two speeds, and let verified data and partners decide each step up.

1. Lead with O&M and technical services for Cleanaway, HiQ, Melbourne Water and REL, through a local consortium, for example with TRILITY or John Holland. Expect the first O&M revenue around 2030–31.
2. Start low-capital moves now: an IWN meter pilot with a regional water corporation, a biosolids proposal to Melbourne Water and an R&D MoU with a Victorian university.
3. Keep technology introduction, co-investment and the coal-site hub as a later track, opened only once policy, regulatory and approval risks are clear. Avoid firm commitments until after the 28 November 2026 state election and the government's response to the inquiry.

90-day actions

1. Commission independent verification of Wanxing and Longfeng emissions in EU format.
2. Prepare an English technical dossier with operating KPIs and case studies.
3. Start NMI R 49 testing for one ultrasonic meter.
4. Brief Invest Victoria and Austrade, and prepare the FIRB application and China-side outbound filing for the Melbourne entity.
5. Approach two consortium partners (a local waste company and an engineering firm) and Melbourne Water.
6. Sign an R&D MoU with a Victorian university.

What CDEG must send us: operating KPIs (availability, kWh/t, cost/t); raw CEMS data that explains the furnace 1 and 4 readings; OIML certificates; meter sales records, since the registration form says 'not yet sold' but lists sales in South Africa, Singapore and Bangladesh; and what the 'One-Four-Five-Five' strategy means.

References

[1] EPA Victoria. (2026). Waste to Energy Scheme licences (updated 27 August 2026). https://www.epa.vic.gov.au/waste-energy-scheme-licences

[2] EPA Victoria. (2026). Waste levy rates. https://www.epa.vic.gov.au/waste-levy

[3] Chengdu Xingrong Environment Co., Ltd. (2026). 2025 annual report (000598.SZ). cninfo. https://www.cninfo.com.cn

[4] Chengdu Xingrong Environment Co., Ltd. (2026). 2026 half-year report (000598.SZ) [Unaudited]. cninfo. https://www.cninfo.com.cn

[5] CDEG–A team. (2026, 15 October). Meeting notes: call with Chengdu Environment Group [Internal record].

[6] Legislative Council Economy and Infrastructure Committee. (2026, August). Inquiry into the development and expansion of waste-to-energy (WtE) infrastructure in Victoria [Report]. Parliament of Victoria. https://www.parliament.vic.gov.au

[7] Sustainability Victoria. (n.d.). Recycling Modernisation Fund and Bioenergy Fund guidelines.

[8] Parliament of Victoria. (2025–2026). Submissions to the Inquiry into the development and expansion of WtE infrastructure in Victoria (627 listed; 565 coded by the team). https://www.parliament.vic.gov.au/get-involved/inquiries/wastetoenergyinquiry/submissions/

[9] DCCEEW. (2026, September). National Waste and Resource Recovery Database 2026 [nwrr-database-2026.xlsx].

[10] Victorian Government. (2024, December). Guidance on the scope of the Waste to Energy Scheme (permitted, banned and exempt waste).

[11] South East Water. (2026, 21 January). South East Water hits 200,000 digital meter milestone. https://southeastwater.com.au/about-us/news/digital-meter-milestone2/

[12] National Measurement Institute. (2026). Certificates of approval 14/3/70 (Sanchuan), 14/3/22 (Ningbo Water Meter), 14/3/72 (Landis+Gyr). https://www.industry.gov.au/sites/default/files/nmi/certificates-approval/2026-01/14-3-70_r3.pdf

[13] Essential Services Commission. (2026). Water Price Review 2028. https://www.esc.vic.gov.au/water/water-prices-tariffs-and-special-drainage/water-price-reviews/water-price-review-2028

[14] Foreign Investment Review Board. (2020, 18 December). Guidance Note 7: Business investments. https://foreigninvestment.gov.au/sites/firb.gov.au/files/guidance-notes/G07-Business.pdf

[15] NexStar 2026. (2026). Collaboration opportunities with Chengdu new energy companies: Chengdu Environment Group profile, strengths and needs [Company document, English and Chinese].

[16] Chengdu Environment Investment Group. (2026). NexStar 2026 enterprise registration form [Company document].

[17] Xizi Clean Energy. (2026, 11 May). All units of Wanxing Phase III complete full-load trial operation. https://c.m.163.com/news/a/KSMA9KOE05568W0A.html

[18] Chengdu Environment Group. (2026, 16 October). Written answers to questions on Australian projects [Company document].

[19] Ministry of Ecology and Environment of China. (2026). MSW incineration plant automatic monitoring data disclosure platform: Chengdu Xingrong Wanxing plant, daily averages of 14 October 2026. https://ljgk.envsc.cn

[20] EPA Victoria. (2026, June). Waste, recycling and resource recovery investment prospectus. https://www.epa.vic.gov.au/sites/default/files/2026-06/waste-recycling-resource-recovery-investment-prospectus.pdf

[21] NSW EPA. (2025). Waste levy rates.

[22] CDEG–A team (Kai). (2026, 14–16 October). Latrobe Valley WtE hub briefing; CDEG Australia market opportunities and entry options; CDEG Australia market research v2; website assumptions.csv [internal].

[23] EPA Victoria. (2025, 1 September; updated 9 December 2025). Cleanaway MERC application withdrawn. https://www.epa.vic.gov.au/cleanaway-merc-application-withdrawn

[24] EQT. (2026, 18 October). EQT agrees to acquire Cleanaway, a leading Australian integrated waste management and resource recovery platform. https://eqtgroup.com/news/eqt-agrees-to-acquire-cleanaway-a-leading-australian-integrated-waste-management-and-resource-recovery-platform-2026-10-18

[25] Victorian Government. (2026). Standardising household recycling across Victoria. https://www.vic.gov.au/standardising-household-recycling-across-victoria

[26] Mornington Peninsula Shire. (2026, 12 March). Council to decide on waste reform implementation. https://www.mornpen.vic.gov.au/The-Shire/News-media-and-publications/News-and-media-releases/Council-to-decide-on-waste-reform-implementation

[27] Essential Services Commission. (2026, September). Regulator releases guidance to assist businesses with water price submissions [Media release]. https://newshub.medianet.com.au/2026/09/regulator-releases-guidance-to-assist-businesses-with-water-price-submissions/171429/

[28] EPA Victoria. (2026). Victoria's Waste to Energy Scheme. https://www.epa.vic.gov.au/victorias-waste-energy-scheme

[29] NDRC and others. (2024). Measures for the administration of infrastructure and public utility concessions (NDRC Order No. 17 of 2024). https://www.gov.cn/gongbao/2024/issue_11346/202405/content_6951524.html

[30] The Paper. (2025, 15 July). How China ran short of waste to burn [in Chinese]. https://www.thepaper.cn/newsDetail_forward_31178082

[31] National Development and Reform Commission. (2012). Notice on improving the pricing policy for waste incineration power generation (Fa Gai Jia Ge [2012] No. 801). https://www.ndrc.gov.cn/xxgk/zcfb/tz/201204/t20120410_964415.html

[32] NDRC, Ministry of Finance and National Energy Administration. (2021). 2021 biomass power project construction work plan (Fa Gai Neng Yuan [2021] No. 1190) [in Chinese]. NDRC interpretation: https://www.ndrc.gov.cn/xxgk/jd/jd/202108/t20210819_1294041.html

[33] Ministry of Ecology and Environment of China. (2019). Regulations on the application and management of automatic monitoring data of MSW incineration power plants (MEE Order No. 10; amended 20 September 2026). https://www.mee.gov.cn/gzk/gz/202112/t20211211_963805.shtml

[34] Ministry of Ecology and Environment of China. (2022, 21 September). Penalty exemptions for MSW incineration power plants, first and second quarters of 2022 [in Chinese]. https://www.mee.gov.cn/ywgz/sthjzf/zfzdyxzcf/202209/t20220921_994584.shtml

[35] Frankston City Council. (2024, 21 August). South East Metropolitan Advanced Waste Processing (SEMAWP). https://www.frankston.vic.gov.au/Council/News-and-media/Latest-News/Media-Releases-2024/South-East-Metropolitan-Advanced-Waste-Processing-SEMAWP

[36] Victorian Government. (2025). Circular Economy (Waste Reduction and Recycling) (Waste to Energy Scheme) Amendment Regulations 2025: Regulatory impact statement. https://www.vic.gov.au/sites/default/files/2025-04/Circular-Economy-%28Waste-Reduction-and-Recylcing%29-%28Waste-to-Energy-Scheme%29-Amendment-Regulations-2025-Regulatory-Impact-Statement-.docx

[37] Harford Consulting for the Smart Energy Council. (2026, July). Waste-to-energy in Australia: energy solution or problem? https://smartenergy.org.au/wp-content/uploads/2026/07/Smart-Energy-Council-and-waste-to-energy-in-Australia-2026.pdf

[38] EPA Victoria. (2017). Energy from waste guideline (Publication 1559.1). https://www.epa.vic.gov.au/15591-energy-waste-guideline

[39] Environmental Justice Australia. (2023). Challenging the Maryvale Mill waste incinerator. https://envirojustice.org.au/legal-work/ending-pollution/ending-toxic-contamination-and-waste/challenging-the-maryvale-mill-waste-incinerator/

[40] EPA Victoria. (2026). Waste to Energy Scheme (updated 27 August 2026). https://www.epa.vic.gov.au/waste-energy-scheme

[41] Xinhua. (2026, 25 May). Urban household waste resource-utilisation rate to exceed 76% by end-2030 [in Chinese]. https://www.news.cn/politics/20260525/8e0c96ba7378447b8b0cc8886512141f/c.html

[42] NDRC and Ministry of Housing and Urban-Rural Development. (2021). 14th Five-Year Plan for urban household waste sorting and treatment facilities (Fa Gai Huan Zi [2021] No. 642) [in Chinese]. https://www.gov.cn/zhengce/zhengceku/2021-05/14/content_5606349.htm

[43] Xinhua Daily Telegraph. (2026, 27 May). An investigation into 'not enough waste to burn' [in Chinese]. https://www.news.cn/politics/20260527/7254958591fd484899511ae57e04bf10/c.html

[44] Ministry of Finance, NDRC and National Energy Administration. (2020). Supplementary notice on promoting the healthy development of non-hydro renewable power (Cai Jian [2020] No. 426). https://www.gov.cn/zhengce/zhengceku/2020-10/21/content_5552978.htm

[45] Wangneng Environment Co., Ltd. (2026). 2026 half-year report: management discussion and analysis (002034.SZ) [in Chinese]. Stockstar. https://wap.stockstar.com/detail/RB2026082800000522

[46] Chengdu Environment Group. (2017, 31 May). Xingrong Environment delegation visits Lahore, Pakistan [WeChat article].

[47] Hatzfeld, S. (2017, 31 May). MBT Jiangsu: high-tech waste processing for China. Recycling Magazine. https://www.recycling-magazine.com/en/20504/mbt-jiangsu-high-tech-waste-processing-china

[48] Jones Day. (2026, June). How long does Australian FIRB approval take? Latest statistics and more streamlining reforms. https://www.jonesday.com/en/insights/2026/06/how-long-does-australian-firb-approval-take-latest-statistics-and-more-streamlining-reforms

[49] Department of Home Affairs. (2023). Security of Critical Infrastructure (Critical infrastructure risk management program) Rules (LIN 23/006). https://www.homeaffairs.gov.au/reports-and-pubs/files/soci-rmp-rules-legislative-instrument-lin-22-018.pdf

[50] National Development and Reform Commission. (2026). Measures for the administration of outbound investment (NDRC Order No. 49 of 2026; effective 1 December 2026, repealing NDRC Order No. 11 of 2017) [in Chinese]. https://www.ndrc.gov.cn/xxgk/zcfb/fzggwl/202610/t20261022_1408298.html

[51] Parliament of Victoria. (2026). 2026 state election. https://www.parliament.vic.gov.au/election

[52] Department of Industry, Science and Resources. (2025, May). Melbourne Energy and Resource Centre AIP plan summary; North Central Review. (2025, September). Cleanaway seeks to double Wollert plant.

[53] ABC News. (2026, 26 August). Parliamentary inquiry into Melbourne waste-to-energy plants prompts calls for tougher regulations. https://www.abc.net.au/news/2026-08-26/vic-report-on-waste-to-energy-calls-for-tougher-regulations/107077314

[54] Australian Tenders. (2025). Melbourne Water: development of a biosolid processing solution. https://www.australiantenders.com.au/tenders/581897/melbourne-water-mw-development-of-a-biosolid-processing-solution

[55] Recovered Energy Laverton. (n.d.). Laverton North Project [Developer website]. https://www.recoveredenergy.com.au/laverton-north-project

[56] RMIT University. (2023, December). Aussie innovation turns waste into energy 'gold' (PYROCO). https://rmit.edu.au/news/all-news/2023/dec/waste-to-bichar

[57] Planning Victoria. (2026). Hazelwood Rehabilitation Project: environment effects statement and draft planning scheme amendment (exhibition 21 September – 2 November 2026).

[58] Premier of Victoria. (n.d.). Loy Yang A transition agreement (closure 2035).

[59] Latrobe Valley Express. (2025, November). New life for Yallourn W; RenewEconomy. (2026, August). EnergyAustralia coal sites.

[60] Opal, Veolia and Masdar Tribe Australia. (2024, 19 August). Maryvale Energy from Waste project wins multi-council waste tenders [Media release]. https://opalanz.com/app/uploads/2024/08/240819-Maryvale-EfW-project-wins-multi-council-waste-tenders_final.pdf

[61] Bureau of Infrastructure and Transport Research Economics. (n.d.). Average road and rail freight rates, 2016 (cited in the team Market Opportunities brief). [To verify]

[62] Opal Australian Paper. (2021). Maryvale EfW fact sheet 11: rail transport of waste from Dynon.

[63] Metro Vancouver. (2025). Waste-to-Energy Facility financial update 2024. https://metrovancouver.org/services/solid-waste/Documents/wte-financial-update-2024.pdf

[64] CDEG–A team. (2026, October). Commercial viability and government support: CDEG entry into Victoria's WtE market [team cost model draft].

[65] South East Water. (2025). Submission to the McCrae inquiry. https://www.mccraeinquiry.vic.gov.au/sites/default/files/2025-09/SEW.0001.0001.5014.pdf

[66] Mail Community. (2026, 21 May). Fifty thousand water meters installed by Yarra Valley Water. https://lilydale.mailcommunity.com.au/news/2026/05/21/fifty-thousand-water-meters-installed-by-yarra-valley-water/

[67] Yicai Global. (2024, 27 August). Huizhong deal to supply an Australian city with smart water meters. https://yicaiglobal.com/news/huizhong-soars-after-chinese-firm-bags-usd140-mln-deal-to-supply-australian-city-with-smart-water-meters

[68] Australian Tenders. (2025). Supply and delivery of water meters (GWW11903). https://www.australiantenders.com.au/tenders/593805/supply-and-delivery-of-water-meters

[69] Bureau of Meteorology. (2026, March). National performance report 2024–25: water and wastewater service providers. https://www.bom.gov.au/water/npr/docs/2024-25/National_Performance_Report_2024-25.pdf

[70] Knox City Council. (2025). Bin audit finds 43% food scraps. https://knox.vic.gov.au/whats-happening/news/knox-bin-audit-finds-43-food-scraps

[71] NAWTEC20. (2012). Characteristics of municipal solid waste in Chengdu [conference paper].

[72] Cleanaway. (2026, July). Bromelton energy-from-waste feedstock report.

[73] Opal Australian Paper. (2021, July). Maryvale Energy from Waste fact sheet 13: waste composition.

[74] Business News WA. (2024–2026). Coverage of the Kwinana sale and East Rockingham receivership.

[75] Department of Industry, Science and Resources. (n.d.). Kwinana Waste to Energy project information.

[76] Clean Energy Finance Corporation. (n.d.). Kwinana and East Rockingham project pages. https://www.cefc.com.au

[77] Macquarie Group. (n.d.). Kwinana WtE media releases.

[78] Bioenergy International. (n.d.). Kwinana waste supply agreements.

[79] Allens. (n.d.). Kwinana WtE project analysis.

[80] ACCC. (2025, 17 July). Acciona – East Rockingham Waste to Energy Project (informal merger review). https://www.accc.gov.au/public-registers/mergers-and-acquisitions-registers/public-informal-merger-reviews-register-2002-25/acciona-east-rockingham-waste-to-energy-project

[81] Columbia University Earth Engineering Center. (2018). Capital cost comparison of waste-to-energy facilities in China and the U.S.

[82] Clean Energy Regulator. (n.d.). Eligible renewable energy sources: biomass-based components of municipal solid waste.

[83] IJGlobal. (n.d.). Kwinana energy-from-waste plant, Australia. https://ijglobal.com/articles/136154/kwinana-energy-from-waste-plant-australia

[84] Infrastructure Pipeline. (n.d.). Kwinana Waste to Energy Project. https://infrastructurepipeline.org/project/kwinana-waste-to-energy-project

[85] ARENA. (n.d.). Kwinana Waste to Energy project.

[86] Parliamentary Budget Office. (2025). Election costing: waste incineration levy.

[87] HiQ. (n.d.). Sunbury Eco-Hub. HiQ EFW Victoria [Developer website].

[88] Parliament of Victoria, Economy and Infrastructure Committee. (2026, 26 May). Transcript of evidence: Opal, OCO Technology and HiQ.

https://www.parliament.vic.gov.au/49f754/contentassets/aa4286fa1a0f4e94a2754449e15bbd7f/2.-final---opal_oco-tech_hiq---26052026.pdf

[89] Infrastructure Partnerships Australia. (2025, 11 April). Melbourne Water energy-from-waste EOI.

[90] VicWater. (n.d.). Working smarter: Barwon Water partners with Iota. https://vicwater.org.au/working-smarter-barwon-water-partners-with-iota-to-accelerate-water-saving-technology-rollout/

[91] Department of Energy, Environment and Climate Action. (2023). Intelligent water networks. https://www.water.vic.gov.au/our-programs/intelligent-water-networks

[92] National Measurement Institute. (2022). NMI R 49-1 Water meters for cold potable water and hot water, Part 1. https://www.industry.gov.au/sites/default/files/2019-05/nmi_r_49-1.pdf

[93] Australian Building Codes Board. (2020). WMTS-530:2020 Water meters with integral shut-off valve. https://codemark.abcb.gov.au

[94] Whirlpool Forums. (2011–2026). Threads on digital and smart water meters (13 threads, 177 posts). https://forums.whirlpool.net.au/

[95] ESD News. (2023). Consortium inks deal with Cobra for Maryvale EfW project. https://esdnews.com.au/consortium-inks-deal-with-cobra-for-maryvale-efw-project/

[96] Cleanaway Waste Management. (2026, 20 August). FY26 results (ASX announcement). https://announcements.asx.com.au/asxpdf/20260820/pdf/072yq2lqg3cv31.pdf

[97] Zawya. (2026, 14 January). Tadweer Group expands global Waste-to-Energy footprint through strategic partnership with Tribe and Cleanaway [Press release]. https://www.zawya.com/en/press-release/companies-news/tadweer-group-expands-global-waste-to-energy-footprint-through-strategic-partnership-with-tribe-and-cleanaway-ewa1f5iv

[98] Infrastructure Investor. (2016). EQT sells waste-to-energy business for €1.4bn. https://www.infrastructureinvestor.com/eqt-sells-waste-to-energy-business-for-e1-4bn/

[99] Veolia. (2025, 12 November). Veolia secures AUD $700m in new contracts to advance technological innovation in water management in Australia [Press release]. https://www.veolia.com/en/our-media/press-releases/veolia-australia-new-contracts-water-management

[100] NSW Department of Planning, Housing and Infrastructure. (2026, 30 July). Veolia Woodlawn ARC [State significant development]. https://www.planning.nsw.gov.au/assess-and-regulate/state-significant-projects/energy-from-waste/veolia-woodlawn-arc

[101] ACCC. (2025). Acciona – East Rockingham Waste to Energy Project, informal merger review register (withdrawn 13 October 2025). https://www.accc.gov.au/public-registers/mergers-registers/public-informal-merger-reviews-register/acciona-east-rockingham-waste-to-energy-project

[102] Waste Management Review. (n.d.). REMONDIS withdraws Swanbank EfW plan. https://wastemanagementreview.com.au/remondis-withdraws-swanbank-efw-plan/

[103] Ipswich City Council. (2022, 1 March). People power puts incinerator on the back burner [News]. https://www.ipswich.qld.gov.au/News-Articles-Folder/2022/People-power-puts-incinerator-on-the-back-burner

[104] TRILITY. (2018). Beijing Enterprises Water Group finalises the acquisition of TRILITY Group. https://trility.com.au/news/beijing-enterprises-water-group-limited-finalises-the-acquisition-of-trility-group/

[105] TRILITY. (n.d.). Adelaide Desalination Plant. https://trility.com.au/project/adelaide-desalination-plant/

[106] Sydney Water. (2022, 12 October). World-class water recycling facility for Western Sydney [Media release]. https://www.sydneywater.com.au/content/dam/sydneywater/media-releases/121022-upper-south-creek-media-release.pdf

[107] Global Construction Review. (2014, 13 December). Chinese construction giant to buy Australia's John Holland in bid for global expansion. https://www.globalconstructionreview.com/chinese-construction-gi0ant-b0uy-aus4tral4ias-john/

[108] Cuthbert, J. (2026, 24 March). Construction giant John Holland hits record $117M loss. Grafa. https://grafa.com/en/news/australia/construction-giant-john-holland-hits-record-117m-loss

[109] Jemena. (2025, May). Pipeline management plan: Jemena gas assets [Safety case]. https://www.jemena.com.au/siteassets/gas/gas-999-pa-hse-002-jemena-safety-case-saop-_-pipeline-management-plan--jemena-gas-assets.22052025.pdf

[110] Jemena. (2015, April). Jemena Electricity Networks (Vic) Ltd regulatory proposal. Australian Energy Regulator. https://www.aer.gov.au/system/files/Jemena%20-%20Regulatory%20proposal%20-%20April%202015.pdf

[111] IPE Real Assets. (2025, 5 November). TIF exits ElectraNet stake in sale to Australian Retirement Trust. https://realassets.ipe.com/news/tif-exits-electranet-stake-in-sale-to-australian-retirement-trust/10133525.article

[112] Utility Magazine. (2016). Treasurer Scott Morrison blocks Chinese investment of Ausgrid. https://utilitymagazine.com.au/treasurer-scott-morrison-blocks-chinese-investment-of-ausgrid/

[113] ABC News. (2025, 23 June). Company in charge of proposed incinerator outside Geelong has direct links to the Chinese government. https://www.abc.net.au/news/2025-06-23/victoria-waste-to-energy-project-lara-chinese-government-links/105442594

[114] City of Greater Geelong. (2025, 29 January; updated 23 March 2026). Advocacy against Lara incinerator continues. https://www.geelongcity.vic.gov.au/whats-happening/news-and-updates/advocacy-against-lara-incinerator-continues

[115] BINDT. (2025, January). Avnet to support South East Water's mass digital meter deployment. https://www.bindt.org/News/january-2025/avnet-to-support-mass-digital-meter-deployment/

[116] CDEG–A team. (2026, October). CDEG Australia market research and entry framework: gates and weighting [Internal working paper].

[117] National Measurement Institute. (2026). About pattern approval. https://www.industry.gov.au/regulations-and-standards/pattern-approval

[118] 清气团 · 固废展望. (2025, 6 May). 澳大利亚垃圾焚烧行业的群雄逐鹿 [Australia's waste-to-energy industry: a contest of contenders]. 中国固废网. https://www.solidwaste.com.cn/news/357711.html

[119] NSW Department of Planning. (2025). Energy from waste: priority infrastructure areas.

[120] NSW EPA. (2026). Waste levy rates; FOGO mandates and rollout.

[121] WA Government. (2026). Waste levy rate schedule.

[122] Queensland Government. (2026). Waste levy rates.

[123] Ministry of Ecology and Environment of China. (2014). GB 18485-2014 Standard for pollution control on the municipal solid waste incineration (amended 2019, MEE Announcement No. 56 of 2019). https://www.mee.gov.cn/ywgz/fgbz/bz/bzwb/gthw/gtfwwrkzbz/201405/t20140530_276307.shtml

[124] European Union. (2010). Directive 2010/75/EU on industrial emissions, Annex VI. https://legislation.gov.uk/eudr/2010/75/annex/VI/part/3/adopted

[125] European Commission. (2019). Implementing Decision (EU) 2019/2010 establishing best available techniques (BAT) conclusions for waste incineration. Official Journal of the European Union, L 312.

Appendix A. Licence-holder matrix

Table 18. Victoria's ten WtE licence holders (first pass)

Holder	Licence	Site and status	Technology	EPC / O&M	CDEG entry point
Cleanaway	Cap 760,000	Wollert MERC; EPA application to be re-filed; EQT takeover pending	Not announced	Not found	O&M if not self-operated or bundled; check with the new owner
HiQ EFW Victoria	Cap 750,000	Bulla; full operation ~late 2030	Kanadevia Inova	Not found	O&M if KVI does not bundle it
Melbourne Water	Cap 450,000	Biosolids EfW; EOI 2025	Open	Open	Biosolids operator: closest fit
Recovered Energy Laverton	Existing-operator; cap 280,000	Laverton North; start not confirmed	Vertical rotating gasification	Not found	Ask who will operate
Liquid Power Co	Cap 50,000	Early development	Gasification + Fischer-Tropsch	Not found	Low fit: below 250 t/d
Zerogen Holdings	Cap 45,000	Wellington precinct	Plasma-assisted gasification	Not found	Low fit: below 250 t/d
Knox Transfer Station	Cap 15,000	To be confirmed	Not found	Not found	Low: small
Paper Australia (Opal), Maryvale	Existing-operator	Pre-construction; no financial close found; A$48.2m federal grant	Not named	Veolia in consortium	Low: operator in place
Visy Industries	Existing-operator	Coolaroo; existing facility	Not checked	Not checked	To research
Great Southern Waste Technologies	Existing-operator	Dandenong South, 100,000 t/yr	Norwegian rights	Not found	Check operator

Source: EPA Victoria licence register and team precedent briefing [1, 23, 24, 60].

Appendix B. Four-state comparison, including waste sorting

Table 19. Victoria against NSW, WA, QLD and SA

Dimension	VIC	NSW	WA	QLD	SA (and ACT, TAS)
Thermal WtE policy	2.5 Mt/yr cap, 94% allocated [1]	Designated precincts only: Parkes, Southern Goulburn–Mulwaree, West Lithgow; Tomago proposed [119]	Permitted; the only state with large plants built [37]	Policy prefers waste-to-fuels; $45m budget support [37]	SA: no large project; ACT bans thermal WtE [37]
Metro landfill levy	$177.19/t (2026–27) [2]	$174.20/t; regional $100.30 (2025–26; 2026–27 to update) [120]	$88/t (2025–26), $90 (2026–27) [121]	$125/t (2025–26), rising to $165 by 2029–30 [122]	SA ~$168 (2025–26) [To verify]; TAS $70.50 [37]
Household sorting	Four bins with standard lids; deadline 1 Jul 2027 (not yet legislated, Mar 2026); FOGO for all households by 2030 [25, 26]	FOGO for households by 2030; large food-waste generators from July 2026 [120]	FOGO rollout in Perth and Peel under the state waste strategy [To verify]	Council-led; no statewide FOGO mandate found [To verify]	Three-bin system common; FOGO widespread [To verify]
Commercial	TEEP test	Energy-from-waste	n/a	n/a	n/a

Dimension	VIC	NSW	WA	QLD	SA (and ACT, TAS)
waste rules	before WtE; deposit containers banned [10]	policy statement; resource-recovery criteria [To verify]			
Waste generated	14.6 Mt (2023), 69% recovered [20]	21.4 Mt (2023–24), 66% recycled; Sydney landfill runs short ~2030 [22]	7.4 Mt (2023–24), 65% recycled [22]	9.81 Mt (2023–24), 57.1% recovered [22]	—
Plants and projects	7 cap licences; Maryvale approved (App. A)	Parkes (Cleanaway, Tadweer, Tribe) [97]; Woodlawn under assessment [100]	Kwinana operating; East Rockingham in receivership [80, 101]	Swanbank withdrawn 2022 [102]	—
Power market	NEM	NEM	WEM (separate grid)	NEM	NEM
Read for CDEG	**Lead market**: most licensed tonnes, organics push, Melbourne network	Second market: precincts and a landfill shortage	Commissioning and O&M for distressed assets	Fastest-rising levy; fuels focus	Small; one water buyer (SA Water)

Sources as cited.

Why sorting matters for the furnace. The more food a state takes out of the red bin, the drier and hotter its residual waste and the fewer tonnes are left (3.3). Victoria and NSW both mandate FOGO by 2030, so residual waste in the two largest markets will move towards the top of CDEG's grate window over the plant life.

Appendix C. Emission standards and heating value

Table 20. Daily emission limits and Wanxing readings (mg/Nm³, 11% O2, dry)

Pollutant	GB 18485-2014 daily	EU IED daily	EU BAT-AEL, new plants	Wanxing, 14 Oct 2026
Dust	20	10	<2–5	0.264
NOx	250	200	50–120	76.29
SO2	80	50	5–30	14.37
HCl	50	10	<2–6	1.961
CO	80	50	10–50	5.824
Mercury	0.05	0.05	0.005–0.02	Pending from CDEG
Dioxins (ng TEQ/Nm³)	0.1	0.1	<0.01–0.04	Pending from CDEG

Sources: [123, 124, 125, 19]. Wanxing values are furnace 1 daily averages; furnace 4 shows identical values, so the data must be verified before use. EU limits are a benchmark here, not a Victorian legal limit [38].

China's national limits are about 1.6 to 8 times the upper EU BAT levels, with HCl furthest apart [123, 125] [Inferred]. Wanxing already has SCR and activated carbon (1.2), so the question for a Victorian plant is less about adding equipment and more about showing that readings stay within BAT levels on hourly and annual averages.

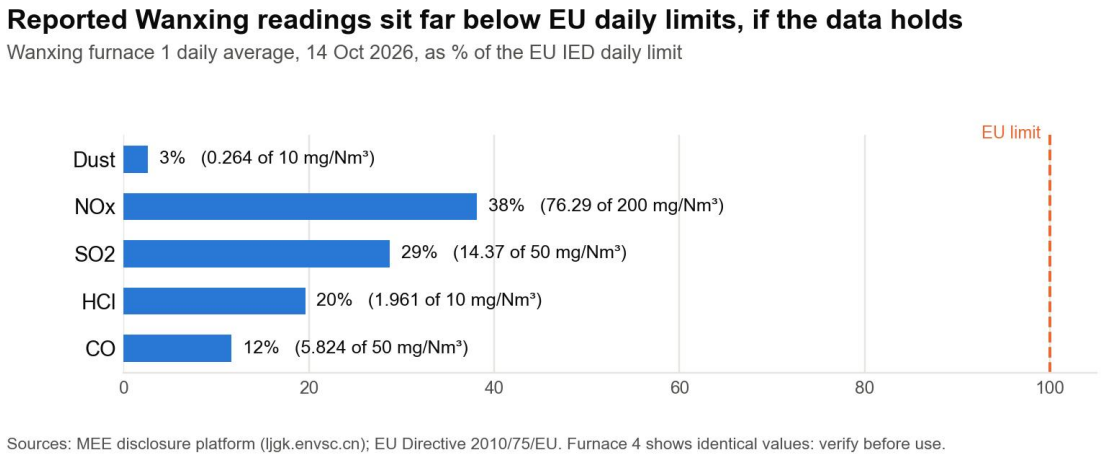


Figure 16. Wanxing daily averages as a share of EU daily limits

Why moisture and heating value matter. Wet food waste lowers the usable heat per kilogram, because energy is spent evaporating water. Dewatering does not add energy; it removes the drag that water puts on the lower heating value and steadies combustion. A furnace is designed to burn waste steadily and completely, recover heat efficiently, limit pollutants and avoid corrosion and slagging. LHV is a key design input for furnace type, throughput and energy recovery, but not the only target. In Victoria the issue reverses: drier, higher-LHV residual waste can push a grate towards its thermal limit.

How a grate plant works. Trucks tip residual waste into a bunker, where a crane mixes it so it burns evenly. The waste moves slowly across a sloping steel grate while air is blown up through it, and the gas must stay above 850°C for at least two seconds to destroy dioxins. The hot gas makes steam in a boiler, and the steam drives a turbine that exports roughly 0.5 to 0.7 MWh per tonne. About 20 to 25% of the input is left as bottom ash, from which metals can be recovered. A smaller amount of fly ash is hazardous and needs special landfill, and selling ash as a product in Victoria needs EPA approval [6].

Appendix D. CDEG asset and KPI sheet

Table 21. CDEG facts used in this report and items still pending

Item	Value	Status
WtE capacity	12,000 t/d (about 6,900 t/d before Wanxing III)	2025 annual report
Wanxing III	5,100 t/d MSW, 800 t/d food waste, 800 t/d sludge; six lines; 72+24-hour full-load trial completed May 2026; on the grid, tipping fee being set; >900 GWh/yr expected	Trial and grid: half-year report; design: supplier release
Minimum economic line	500 t/d; below 250 t/d too small	CDEG answer
Grate design	Design 9,000 kJ/kg; range 4,600–14,000 kJ/kg; load 60–110%	CDEG answer
WtE revenue 2025	RMB 752m; gross margin 45.6%; 1,020 GWh sold	2025 annual report
Meters	Made by sister company Huijin; RMB 85.75m bought by Xingrong in 2025	2025 annual report
Operating KPIs	Availability, kWh/t, cost/t	Pending from CDEG
Raw CEMS data	Hourly and annual averages; dioxin tests	Pending from CDEG
OIML certificates and meter sales	Certificates; overseas customers	Pending from CDEG

Appendix E. Model notes (Models 1 and 3)

Model 1. Cost and O&M contract calculator

Question. What gate fee does a Victorian WtE plant need to break even, how does that compare with landfill, and what is an O&M contract worth to the operator and to the owner?

Table 22. Inputs (base case)

Input	Base	Range tested	Basis
Plant	2 × 500 t/d = 365,000 t/yr design	—	CDEG minimum line [5]
Utilisation	85%	70–95%	[Assumption]; grate plants usually run 85–92%
Capex	$1,600 per t/yr ($584m)	$1,200–2,750	East Rockingham ~$1,700; Maryvale ~$1,850; Kwinana ~$2,750 final [37, 22, 74]
Cost of capital, life	8%, 25 years (capital recovery factor 0.0937)	—	[Assumption]
Fixed O&M	2.5% of capex a year	—	[Assumption] [64]
Variable cost	$40/t (reagents, ash, water)	—	[Assumption] [64]
Power	0.6 MWh/t at $90/MWh	$60–90	[Assumption]; Harford's model implies ~0.8 MWh/t [37]
LGCs	50% biogenic × 0.6 MWh × $15	—	Biomass fraction only [82]
Landfill benchmark	$247/t = levy $177.19 + gate $70	—	[2]; gate fee [Assumption]
WtE levy	$0	$20/t	Federal Greens 2025 proposal [64]

Formulas. Cost per tonne = (capex × CRF + capex × 2.5%) ÷ tonnes + variable cost. Energy income per tonne = 0.6 × power price + 0.5 × 0.6 × LGC price. Break-even gate fee = cost per tonne − energy income + any WtE levy. O&M value per tonne = fixed O&M per tonne + variable cost. Owner value of a 1% opex saving = 1% × O&M value × tonnes × annuity factor (10.67 at 8% over 25 years).

Table 23. Outputs

Scenario	Tonnes/yr	Cost/t	Energy income/t	Break-even/t	vs landfill $247
AU capex, 70%	255,500	311	58.5	253	−6 (not viable)
AU capex, 85% (base)	310,250	263	58.5	205	+42
AU capex, 95%	346,750	240	58.5	181	+66
Capex −25%, 85%	310,250	208	58.5	149	+98
Capex $1,850, 85%	310,250	298	58.5	240	+7
Kwinana final capex (~$2,750/t), 85%	310,250	424	58.5	366	−119 (not viable)
Power $60, 85%	310,250	263	40.5	223	+24
$20 WtE levy, 85%	310,250	263	58.5	225	+22

Contract outputs (base): O&M value $87/t, or $27.0m a year; operator margin at 10–15% is $2.7–4.1m a year. A 1% opex saving is worth $0.27m a year to the owner, or $2.9m in present value. Utilisation at which break-even equals landfill: about 72%.

Heating-value estimate (3.3). Victorian red bin without FOGO, by weight: food 35%, garden 7%, paper 10%, plastics 15% (from [73]); textiles 5%, nappies 8%, wood 3%, glass 3%, metals 3%, other 11% (our allocation of the remainder) [Assumption]. Heating values as received (MJ/kg): food 4, garden 6, paper 12, plastics 32, textiles 15, nappies 8, wood 15, glass and metals 0, other 5. The method reproduces Bromelton's 10.6 MJ/kg design point [72].

Limits. This is a comparison model, not a bankable forecast. It leaves out metals and product revenue, tax, inflation, construction delay, a front-end sorting plant, levy on landfilled ash and the extra cost of EU-standard gas cleaning. Each would raise costs. It also assumes tonnes are not limited by heat input (see 3.3). Replace the inputs with CDEG's cost data when they arrive. The interactive version on the team site adds sliders for each input [To verify] (still to be linked to these inputs).

Model 3. Siting screen

Question. Which sites work for a CDEG-operated plant under Australian conditions, rather than under China's 50 km rule?

Method. Each site is scored 1–5 on six weighted criteria, using public data. The team map (vic_waste_rail_map) shows the layers: licence and coal-plant sites, transfer stations, rail, terminal stations and water.

- haul cost to Melbourne's waste (30%), using the $/t estimates in 3.1e
- grid connection (15%)
- buffer to housing (20%), given that there is no minimum buffer [6]
- approvals in hand (15%)
- water and steam users (10%)
- rail access (10%)

Table 24. Siting screen (first pass)

Site	Haul 30%	Grid 15%	Buffer 20%	Approvals 15%	Water/steam 10%	Rail 10%	Score /5
Melbourne Water ETP, Bangholme (~30 km)	5	3	4	3	5	2	3.9
Melbourne Water WTP, Werribee (~35 km)	4	3	5	3	5	3	3.9
Laverton North, REL (~17 km)	5	4	3	4	3	3	3.9
Wollert, Cleanaway (~25 km)	5	3	2	2	2	2	3.05
Bulla, HiQ (~25 km)	5	3	1	1	2	1	2.6
Hazelwood, Latrobe (~150 km)	2	5	3	2	4	4	3.05
Yallourn, Latrobe (~150 km, after 2028)	2	5	4	1	4	3	3.0

Scores are team judgement [Inferred]; distances are straight-line and must be checked by road [22]. Weights can be changed in the model.

Reading. Melbourne Water's treatment plants and Laverton North tie at 3.9, and the ranking among them depends on the weights. They are close to the waste, have room for buffers, and use the treatment plants' own need for heat to dry sludge. The Latrobe sites lead on grid and rail, but haul cost and approval timing hold them back. They become competitive if a metro project is blocked and moves its licence, or if rail haul comes in below the estimates in 3.1e.

Appendix F. Glossary

Table 25. Chinese–English terms

English	Chinese	Meaning
Waste-to-energy (WtE)	垃圾焚烧发电	Burning residual waste to make power and heat
Cap licence	配额许可	Victorian licence for thermal WtE tonnes under the 2.5 Mt/yr cap
Operation and maintenance (O&M)	运营管理	Running a plant for its owner under contract
Lower heating value (LHV)	低位热值	Usable heat per kilogram of waste
Continuous emissions monitoring (CEMS)	在线监测	Real-time stack emission measurement
FOGO	食物和园林有机垃圾分类	Food organics and garden organics kerbside collection
Pattern approval	计量型式批准	NMI approval for a meter type used in trade
BOT / BOO	建设-运营-移交 / 建设-拥有-运营	Concession models used in China
SASAC	国资委	State-owned assets supervisor
FIRB	外国投资审查委员会	Reviews foreign investment; an SOE needs approval to start any Australian business
SOCI Act	关键基础设施安全法	Security duties for owners of critical water and energy assets
Outbound investment filing (ODI)	境外投资备案	China-side filing with the provincial DRC before investing abroad
Gate fee	进场处理费	Fee a waste-to-energy plant charges per tonne of waste received

Appendix G. Evidence labels

Table 26. Labels used in this report

Label	Meaning
[Official]	Government, regulator or parliament record
[Industry]	Industry report, company announcement or media
[Company]	CDEG's own material, not independently verified
[Inferred]	Our own calculation or judgement from the sources
[Assumption]	A planning value chosen by the team, to be replaced with data
[To verify]	Incomplete or conflicting; needs checking